

Numerical Methods for Partial Differential Equations

Lecture Notes 2023/24

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Chapter 1

Introduction: Some Basic Concepts

In this course, we introduce numerical methods for the solution of **Partial Differential Equations** (PDEs), with focus on the **Finite Element** (FE) method and the use of the computer for the construction of the PDEs numerical solution. We will consider the numerical approximation of elliptic and parabolic PDEs by considering their variational formulation, Galerkin and FE approximations in 1D/2D/3D, the theoretical properties and practical use of the methods, algorithmic aspects, and interpretation of the numerical results. Advanced topics include the approximation of saddle-point PDEs (Stokes equations), vectorial, nonlinear, and multiphysics differential problems, domain decomposition methods exploiting the properties of the PDEs, and the introduction to parallel computing for the FE method, i.e. in the **High Performance Computing** (HPC) framework. The course will feature the use of the *deal.II* software library [1, 8], a C++ open source FE library, and *ParaView* [9] for the visualization of numerical solution and scientific computing data.

Most of the content of these lecture notes refers to [3], which covers most of the topics of the course. Additional material can be found in [2, 4, 5, 6, 7].

1.1 Mathematical Models and Scientific Computing

A **mathematical model** is a set of (algebraic or differential) equations that is able to represent the features of a complex system or process. Models are developed to describe, forecast, and control the behavior or evolution of such systems.

Mathematical modeling assumes nowadays a crucial role in the description of several phenomena in Engineering and applied Sciences. The computational power available enables a widespread use of numerical methods for simulating mathematical models. In industry, for example, computational tools are a key factor for the design of products and technological developments.

Physics-based models are those mathematical models that are derived from physical principles – like conservation laws of mass, momentum, energy, etc. – and that encode natural laws leading to (differential) equations whose solutions are often represented in the form of functions. However, the analytical solution of such models is rarely available in closed form, for which numerical approximation methods are instead employed. **Numerical modeling** indicates sets of numerical methods that determine an approximate solution of the original (often infinite-dimensional) mathematical model, by turning it into a discrete problem (algebraic, finite-dimensional), whose dimension (size) is typically very large. **Scientific Computing** is a branch of Mathematics that numerically solves (differential) mathematical models by building approximate solutions through the use of a **calculator**. For numerical models of large size, parallel architectures for calculators and the HPC framework are typically used.

Mathematical models are conventionally used altogether with theoretical (mathematical) models and experimental tests. In several cases, however, theoretical models are not available – like in Computational Medicine – or experimental tests are not meaningful or cannot be performed (for example, for nuclear testing, extreme natural events as earthquakes, etc.). Physics-based models have witnessed an increasing

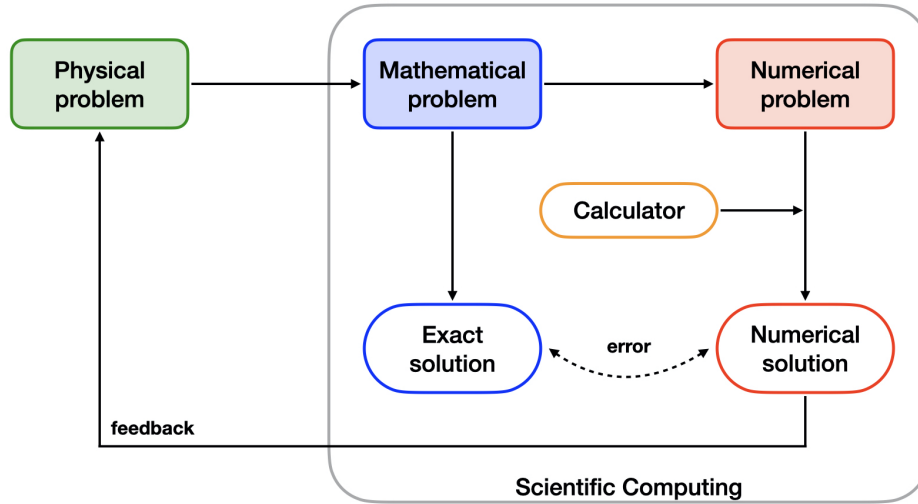


Figure 1.1: Scientific Computing: physical, mathematical, and numerical problems and solutions

role in the modern society in virtue of the massive developments of Scientific Computing and computational tools starting from the 1950s and 1960s. More recently, we entered in a new era characterized by Data Science. Indeed, a large amount of **data** is becoming available from multiple sources nowadays (digital measures, experiments, etc.). Data-driven models are those mathematical models built from meaningful data that do not rely on physical principles – because the latter are not available or are not reliable – and whose construction calls for statistical learning methods.

In any case, data are crucial for physics-based models: as a matter of fact, despite the different paradigms, data-driven and physics-based models can be and must be used in a synergistic way. Physics-based models require input data (in the form of physical and geometrical parameters for differential equations) and **validation** through experimental tests and measurements. More recently, new ways of balancing data-driven and physics-based models have emerged and are increasingly used as numerical models. An example, is represented by Scientific Machine Learning, an evolution of Scientific Computing that exploits statistical learning based on Machine and Deep Learning algorithms. This can be used to accelerate or empower standard numerical methods in Scientific Computing, to reduce complexity and computational costs of the numerical simulations, or to build **digital twins**. A digital twin is a virtual representation of a real physical system or process that serves as its digital counterpart for practical uses, such as simulation, testing, monitoring, maintenance, prediction, and forecasting.

Physics-based mathematical models (**mathematical problems**) are a fundamental pillar in the understanding and prediction of several physical phenomena and processes (**physical problems**). However, these mathematical models lead to problems that can rarely be solved analytically, or in an exact way (**exact solution**), especially for PDEs: with only a few exceptions, it is not possible to write their solution explicitly.

Numerical methods and numerical approximation techniques (**numerical problems**) serve the purpose to determine an **approximate solution** of a mathematical model. When the calculator is used to determine such approximate solution, the latter is called **numerical solution**; see the sketch reported in Fig. 1.1. The course aims at introducing numerical methods for the approximation of PDEs, in particular by means of the FE method, and the use of the calculator and HPC resources for calculating their approximate solution.

1.2 Differential Models and Partial Differential Equations

A differential equation (model) is an equation that involves one or more derivatives of an unknown function. In an **Ordinary Differential Equation** (ODE), every derivative of the unknown solution is with respect to a single independent variable. If instead, derivatives are partial, then we have a **Partial Differential**

Equation (PDE). The nature of the differential equations also depends on **conditions** and their type. The differential model can be indicated in an abstract manner as follows:

$$\mathcal{P}(u; g) = 0 \quad \text{differential equation (mathematical problem)}$$

In the former, $\mathcal{P}$ generically indicates the model, u is the exact solution – a function of one or more independent variables (space and/or time variables) – and g indicates the data.

We report in the following typical differential models, involving ODEs and PDEs; we consider specifically linear models for the PDEs with a scalar unknown solution. For the examples concerning PDEs, we limit to consider elliptic and parabolic PDEs, which are the focus of this course. More details on PDEs will be provided in the following.

ODEs

ODEs are also known as initial values problems. A first order ODE, a Cauchy problem, is a differential problem, whose solution $u = u(t)$ is a function of a single independent variable t , often interpreted as time. A single condition is assigned on the solution, at a point (usually, the left end of the integration interval). Its form is the following: find $u : I \subset \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{du}{dt}(t) = f(t, u(t)) & t \in I, \\ u(t_0) = u_0, \end{cases} \quad (1.1)$$

where $I = (t_0, t_f] \subset \mathbb{R}$ is a time interval, u_0 is the initial value assigned at $t = t_0$, and $f : I \times \mathbb{R} \rightarrow \mathbb{R}$. The equation describes the evolution of a scalar quantity u over time t , without distribution in space.

In vectorial problems, the unknown is a vector-valued function $\mathbf{u} = \mathbf{u}(t)$, where $\mathbf{u} = (u_1, \dots, u_m) \in \mathbb{R}^m$, with $m \geq 1$. The first order Cauchy problem reads: find $\mathbf{u} : I \subset \mathbb{R} \rightarrow \mathbb{R}^m$ such that

$$\begin{cases} \frac{d\mathbf{u}}{dt}(t) = \mathbf{f}(t, \mathbf{u}(t)) & t \in I, \\ \mathbf{u}(t_0) = \mathbf{u}_0, \end{cases}$$

where $\mathbf{u}_0 \in \mathbb{R}^m$ is the initial datum and $\mathbf{f} : I \times \mathbb{R}^m \rightarrow \mathbb{R}^m$.

A second order Cauchy problem sees second order time derivatives and two initial conditions. Its reads as: find $u : I \subset \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{d^2u}{dt^2}(t) = f\left(t, u(t), \frac{du}{dt}(t)\right) & t \in I, \\ \frac{du}{dt}(t_0) = v_0, \\ u(t_0) = u_0, \end{cases} \quad (1.2)$$

where the initial data are u_0 and v_0 , while $f : I \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$.

PDE, boundary value problem in 1D

It is a stationary differential model with a single independent variable x , representing the space coordinate in an interval $\Omega = (a, b) \subset \mathbb{R}$ (1D). The problem involves second order derivatives of the unknown solution $u = u(x)$ with respect to x . The value of u , or the value of its first derivative, is set at the two boundaries of the domain (interval) Ω , that is at $x = a$ and $x = b$ (the domain boundary is $\partial\Omega = \{a, b\}$).

Let us consider the following *Poisson* problem with (homogeneous) Dirichlet boundary conditions: find $u : \Omega \subset \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\frac{d^2u}{dx^2}(x) = f(x) & x \in \Omega = (a, b), \\ u(a) = u(b) = 0. \end{cases} \quad (1.3)$$

This equation models a stationary phenomenon – the time variable does not appear in fact – and represents a diffusion model. For example, this models the diffusion of a pollutant along a 1D channel $\Omega = (a, b)$ or the vertical displacement of an *elastic thread* fixed at its ends. In the first case, $f = f(x)$ indicates the source of the pollutant along the flow, while in the second case, f is the transverse force acting on the elastic thread, in the hypothesis of negligible mass and small displacements of the thread.

We remark that the boundary value problem in 1D is a particular case of PDEs, even if it involves only derivatives with respect to a single independent variable x . Indeed, even if apparently similar to a second order ODE, the boundary value problem is in reality substantially different from an ODE: in Eq. (1.2), two conditions are set at $t = t_0$, while in Eq. (1.3), one condition is set at $x = a$ and the other one at $x = b$. The conditions in the boundary value problem determine to the so-called global nature of the model.

PDE, initial and boundary value problem in 1D

These problems concern equations that depend on space and time: the unknown solution $u = u(x, t)$ both depends on the space coordinate $x \in \Omega \subset \mathbb{R}$ in 1D and the time variable $t \in I \subset \mathbb{R}$. In this case, the initial conditions at $t = 0$ must be prescribed, as well as the boundary conditions at the ends of the interval in 1D.

The *heat equation* – also known as *diffusion equation* – with Dirichlet boundary conditions assumes the following form: find $u : \Omega \times I \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{\partial u}{\partial t}(x, t) - \mu \frac{\partial^2 u}{\partial x^2}(x, t) = f(x, t) & x \in \Omega = (a, b), t \in I, \\ u(a, t) = u(b, t) = 0 & t \in I, \\ u(x, t_0) = u_0(x) & x \in \Omega = (a, b). \end{cases} \quad (1.4)$$

For example, the unknown function $u(x, t)$ describes the temperature in a point $x \in \Omega = (a, b)$ and time $t \in I$ of a metallic bar covering the space interval Ω . The diffusion coefficient μ represents the thermal response of the material and it is related to its thermal conductivity. The Dirichlet boundary conditions express the fact that the ends of the bar are kept at a reference temperature (zero degrees in this case), while at the time $t = t_0$, the temperature is assigned in each point $x \in \Omega$ through the initial function $u_0(x)$. Finally, the bar is subjected to a heat source of linear density $f(x, t)$. Another example is the wave equation, which involves second order derivatives with respect to both space and time variables; the latter is however an hyperbolic PDE.

PDE, boundary value problem in multidimensional domains $\Omega \subset \mathbb{R}^d$, with $d = 2, 3$

Problem (1.3) can be extended in multidimensional domains $\Omega \subset \mathbb{R}^d$, with $d = 2, 3$; the solution is $u = u(\mathbf{x})$, where $\mathbf{x} = (x_1, \dots, x_d)^T \in \mathbb{R}^d$. This leads to the following Poisson problem with (homogeneous) Dirichlet boundary conditions: find $u : \Omega \subset \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\Delta u = f & \text{in } \Omega \text{ (i.e. } \mathbf{x} \in \Omega), \\ u = 0 & \text{on } \partial\Omega \text{ (i.e. } \mathbf{x} \in \partial\Omega), \end{cases} \quad (1.5)$$

where

$$\Delta u(\mathbf{x}) := \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2}(\mathbf{x})$$

is the *Laplace operator*, the domain $\Omega \subset \mathbb{R}^d$ is endowed with boundary $\partial\Omega$, and $f = f(\mathbf{x})$ is the external forcing term. This equation is used for example to model the vertical displacement of an elastic membrane fixed at the boundaries.

PDE, initial and boundary value problem in multidimensional domains $\Omega \subset \mathbb{R}^d$, with $d = 2, 3$

The multidimensional counterpart of the heat equation Eq. (1.4) reads: find $u : \Omega \times I \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{\partial u}{\partial t} - \mu \Delta u = f & \mathbf{x} \in \Omega, t \in I, \\ u(\mathbf{x}, t) = 0 & \mathbf{x} \in \partial\Omega, t \in I, \\ u(\mathbf{x}, t_0) = u_0(\mathbf{x}) & \mathbf{x} \in \Omega, \end{cases} \quad (1.6)$$

where u_0 is the initial datum. Here, the solution is $u = u(\mathbf{x}, t)$.

Classification of PDEs

As anticipated, a PDE is a relationship among the partial derivatives of a function $u = u(\mathbf{x}, t)$, the PDE solution, which typically depends on the spatial coordinates $\mathbf{x} = (x_1, \dots, x_d)^T \in \mathbb{R}^d$ – if the problem is defined in a spatial domain $\Omega \subset \mathbb{R}^d$ – and on the time variable t . We try now to provide a general framework for PDEs. In this respect, a PDE can be written as:

$$\mathcal{P} \left(u, \frac{\partial u}{\partial t}, \frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_d}, \dots, \frac{\partial^{p_1+\dots+p_d+p_t} u}{\partial x_1^{p_1} \dots \partial x_d^{p_d} \partial t^{p_t}}, \mathbf{x}, t; g \right) = 0 \quad (1.7)$$

where $p_1, \dots, p_d, p_t \in \mathbb{N}$ and g stands for the data. The PDE *order* is the maximum order of derivation that appears in $\mathcal{P}$, that is $q = p_1 + \dots + p_d + p_t$. The PDE is *linear* if $\mathcal{P}$ linearly depends on u and its derivatives.

PDEs can be classified as belonging to three groups: *elliptic*, *parabolic*, and *hyperbolic* PDEs. Let us focus on linear PDEs of order $q = 2$ with constant coefficients, such that the general formulation of Eq. (1.7) reads:

$$\mathcal{L} u = g,$$

where $\mathcal{L}$ is a second order, linear differential operator. When only two independent variables x_1 and x_2 are considered, the operator $\mathcal{L}$ applied to the function u reads:

$$\mathcal{L} u = A \frac{\partial^2 u}{\partial x_1^2} + B \frac{\partial^2 u}{\partial x_1 \partial x_2} + C \frac{\partial^2 u}{\partial x_2^2} + D \frac{\partial u}{\partial x_1} + E \frac{\partial u}{\partial x_2} + F u,$$

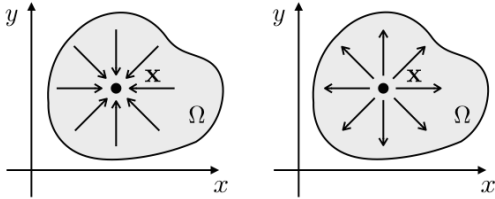
for some constant coefficients A, B, C, D, E, F , and $G \in \mathbb{R}$. The independent variables x_1 and x_2 can represent the space coordinates $x_1 = x$ e $x_2 = y$ (if $d = 2$), as well as the space coordinate $x_1 = x$ (if $d = 1$) and the time variable $x_2 = t$. After introducing the discriminant $\Delta := B^2 - 4AC$, the PDE is classified as: (i) *elliptic*, if $\Delta < 0$; (ii) *parabolic*, if $\Delta = 0$; and (iii) *hyperbolic*, if $\Delta > 0$.

Other than the classification, the different nature of the PDE impacts on: the type and amount of data to prescribe as boundary and initial conditions to ensure the well-posedness of the problem (existence and uniqueness of the solution); the phenomena that can be described by the PDE and the information that encapsulates. Elliptic PDEs typically describe stationary phenomena, without time evolution of quantities. Hyperbolic PDEs describe wave propagation phenomena with finite velocity of propagation; instead, for parabolic PDEs, information is propagated at infinite velocity.

Example 1.1. The following Laplace equation in a 2D domain – a particular case of the Poisson equation (1.5) – is elliptic:

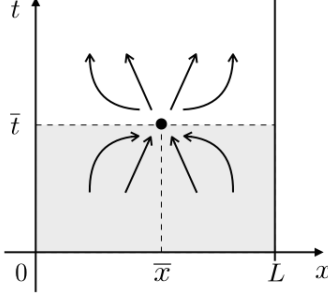
$$\begin{cases} -\Delta u = 0 & \text{in } \Omega \subset \mathbb{R}^2, \\ u = 0 & \text{su } \partial\Omega. \end{cases}$$

Here, $x_1 = x$, $x_2 = y$, and $\mathcal{L}u = -\Delta u = -\frac{\partial^2 u}{\partial x_1^2} - \frac{\partial^2 u}{\partial x_2^2}$, for which $A = C = -1$, $B = 0$, $D = E = F = 0$, and $\Delta = -4 < 0$.



The solution u of this PDE in a point $\mathbf{x} \in \Omega$ is affected by the value of the solution in every point $\mathbf{y} \in \Omega$; moreover, the solution $u(\mathbf{x})$ is able to impact on the solution u in every point $\mathbf{y} \in \Omega$.

Example 1.2. The heat equation (1.4) is parabolic; here, we set $a = 0$, $b = L$, and $t_0 = 0$ for simplicity. Indeed, we have, $x_1 = x$, $x_2 = t$, and $\mathcal{L}u = \frac{\partial u}{\partial t} - \mu \frac{\partial^2 u}{\partial x^2}$, for which $A = -\mu$, $B = C = D = F = 0$, $E = 1$, and $\Delta = 0$.

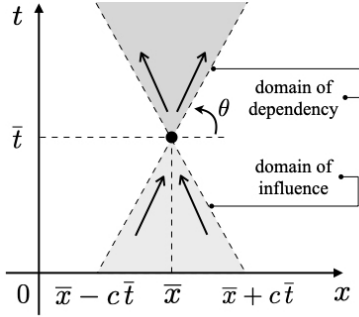


The solution u in $x = \bar{x}$ and $t = \bar{t}$ is affected by the solution $u(x, t)$ for all $x \in [0, L]$ and for every $t \in [0, \bar{t}]$. Moreover, the solution u in $x = \bar{x}$ and $t = \bar{t}$ affects the solution $u(x, t)$ in every point $x \in [0, L]$ and for every time $t > \bar{t}$. This PDE respects the arrow of time (causality principle): the value of the solution u at time $t = \bar{t}$ is independent of any variation of the data occurring at times $t > \bar{t}$.

Example 1.3. The wave equation, defined in a suitable spatio-temporal domain, reads:

$$\mathcal{L}u = \frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0;$$

we assume that the former is endowed with initial and boundary conditions. Here, we set $x_1 = x$ and $x_2 = t$, for which $A = -c^2$, $B = 0$, $C = 1$ and $D = E = F = 0$, thus yielding $\Delta = 4c^2 > 0$. Therefore, the PDE is hyperbolic and $c > 0$ represents the (finite) velocity of wave propagation.



The solution u in $x = \bar{x}$ and $t = \bar{t}$ is affected only by the solution u in the domain of influence, the cone defined by $(\bar{x}, \bar{t})$, $(\bar{x} - c\bar{t}, 0)$, $(\bar{x} + c\bar{t}, 0)$. Moreover, the solution u in $x = \bar{x}$ and $t = \bar{t}$ affects the solution u at times $t > \bar{t}$ in the spatio-temporal subset defined by $\bar{x} - ct \leq x \leq \bar{x} + ct$; the latter is called domain of dependence of $(\bar{x}, \bar{t})$.

1.3 Numerical Methods

As anticipated, in general, we cannot analytically solve a PDE in most of the cases of practical interest. This highlights the importance of using **numerical methods** that allow to construct an approximation u_h of the exact solution u , for which the corresponding error $(u - u_h)$ can be quantified and/or estimated. Here, the subscript h indicates a discretization parameter that characterizes the numerical approximation. Conventionally, the smaller is h , the better is the approximation of u made by u_h ; that is, the error $(u - u_h)$ tends to zero as h gets smaller and smaller. In general, we can provide the following representation:

$\mathcal{P}(u; g) = 0$	PDE (mathematical problem)
$\downarrow$	<i>numerical method</i>
$\mathcal{P}_h(u_h; g_h) = 0$	approximate PDE (numerical problem)

where g_h is an approximation of the data g , while $\mathcal{P}_h$ is a characterization of the approximate problem. In this course, we will specifically introduce the FE method to build the numerical approximation of PDEs.

As example, we present in the following a numerical method based in the *finite differences* for the approximation of the boundary value problem (1.3) for $\Omega = (0, 1)$. Let us assume that the differential equation must be satisfied in every node x_j internal to the domain $\Omega = (0, 1)$, obtained by subdividing

$[0, 1]$ through $N + 1$ nodes $x_j = jh$, for $j = 0, \dots, N + 1$, into subintervals of size $h = \frac{1}{N + 1}$. In practice, we ask that

$$-\frac{d^2u}{dx^2}(x_j) = f(x_j) \quad \text{for all } j = 1, \dots, N.$$

We can approximate this set of N equations by replacing the second derivative with the following *finite difference* at a general point $\bar{x} \in \Omega$:

$$\delta^2 u(\bar{x}) = \frac{u(\bar{x} + h) - 2u(\bar{x}) + u(\bar{x} - h)}{h^2},$$

for $h > 0$. If $u : \Omega \rightarrow \mathbb{R}$ is “sufficiently” regular in a neighborhood of $\bar{x} \in (0, 1)$, then $\delta^2 u(\bar{x})$ approximates $\frac{d^2u}{dx^2}(\bar{x})$ with accuracy of order 2 with respect to h , that is the error $\left| \frac{d^2u}{dx^2}(\bar{x}) - \delta^2 u(\bar{x}) \right|$ goes as h^2 .

For the problem (1.3), instead of searching a function $u(x)$ that satisfies this boundary value problem, we can look for a set of N real values $\{u_j\}_{j=1}^N$ such that

$$\begin{cases} -\frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} = f(x_j) & \text{for all } j = 1, \dots, N, \\ u_0 = u_{N+1} = 0, \end{cases}$$

by means of the finite difference method. In this manner, u_j represents an *approximation* of $u(x_j)$. The former system of equations leads to a *linear system*

$$A\mathbf{u} = \mathbf{f},$$

where $\mathbf{f} = (f(x_1), f(x_2), \dots, f(x_{N-1}), f(x_N))^T$, $\mathbf{u} = (u_1, \dots, u_N)^T$ is the vector of unknowns, and $A \in \mathbb{R}^{N \times N}$ is the following tridiagonal matrix

$$A = \frac{1}{h^2} \text{tridiag}(-1, 2, 1) = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & \ddots & & \vdots \\ 0 & \ddots & \ddots & -1 & 0 \\ \vdots & & -1 & 2 & -1 \\ 0 & \dots & 0 & -1 & 2 \end{bmatrix},$$

which is symmetric and positive definite. As N is typically large, a computer is used to solve the linear system $A\mathbf{u} = \mathbf{f}$ by means of suitable method; for a tridiagonal matrix, the Thomas algorithm – a direct method – is usually used.

If $f \in C^2([0, 1])$, the following error estimate can be derived

$$\max_{j=0, \dots, N+1} |u(x_j) - u_j| \leq \frac{h^2}{96} \max_{x \in [0, 1]} \left| \frac{d^2 f}{dx^2}(x) \right|.$$

We deduce that the error decays with a rate proportional to h^2 (the smaller is $h < 1$, the smaller is the error).

The former example based on finite differences illustrates that the numerical approximation of a PDE calls in reality for multiple numerical methods. Focus of this course is the FE method for the numerical approximation of PDEs, leading to numerical solutions u_h to be determined by means of the computer.

1.4 From the Mathematical Problem to the Numerical Problem

We introduce a general framework for the analysis of a numerical method. We also introduce key concepts such as the well-posedness of a problem, the consistency, stability, and convergence of a numerical method, as well as the conditioning number.

1.4.1 The mathematical problem

Let us consider a *physical problem* (PP) endowed with a *physical solution*, let say u_{ph} , and dependent on data indicated with g . The *mathematical problem* (MP) is represented by the mathematical formulation of the PP and has a *mathematical solution* u . We indicate the PM as:

$$\mathcal{P}(u; g) = 0, \quad (1.8)$$

where $u \in \mathcal{U}$ and $g \in \mathcal{G}$, with $\mathcal{U}$ e $\mathcal{G}$ two suitable sets or spaces; $\mathcal{G}$ is the set or space of admissible data. The error between the physical and mathematical solutions is called *model error*, say $e_m := u_{ph} - u$. This source of error takes into account all those characteristics of the PP that are not represented or captured by the MP.

Example 1.4. Let us consider as PP the elastic thread of Eqs. (1.3) and (??), whose mathematical solution u is a function representing the vertical displacement. Then, the MP can be written as

$$\mathcal{P}(u; g) = u - x \int_0^1 (1-s)f(s)ds + \int_0^x (x-s)f(s)ds = 0,$$

where data are $g = \{(0, 1), 0, 0, f\}$, representing the domain Ω , the homogeneous Dirichlet data, and the forcing term f , respectively. Here, $\mathcal{G} = \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R} \times C^0([0, 1])$ and $\mathcal{U} = C^2([0, 1])$.

Before solving a MP, it is required to ensure that this is *well-posed*.

Definition 1.1. The MP $\mathcal{P}(u; g) = 0$ is well-posed (*stable*) if and only if there exists a unique solution $u \in \mathcal{U}$ that continuously depend on the data $g \in \mathcal{G}$.

$\mathcal{G}$ is the set of admissible data, i.e. those for which the MP (1.8) admits a unique solution. Roughly speaking, “continuous dependence on data” means that “small” perturbations on data $g \in \mathcal{G}$ lead to “small” changes on the solution $u \in \mathcal{U}$ of the MP.

MPs that are well-posed can exhibit “large” variations of the solution u even for “small” variations of the data value d . A measure of this sensitivity is given by the condition number of the MP.

1.4.2 The numerical problem

The *numerical problem* (NP) is an approximation of the MP (1.8). We indicate its *numerical solution* as u_h , where h stands as a suitable *discretization parameter*¹. We state the NP as:

$$\mathcal{P}_h(u_h; g_h) = 0, \quad (1.9)$$

where $u_h \in \mathcal{U}_h$ and $g_h \in \mathcal{G}_h$, with $\mathcal{U}_h$ and $\mathcal{G}_h$ two suitable sets or spaces; g_h is the representation of the data in the NP. The error between the mathematical and numerical solutions is called *truncation error*, say $e_h := u - u_h$, as depicted in Fig. 1.2. This can be regarded as the error that stems from the discretization of the MP.

Example 1.5. Let us consider the following MP: $\mathcal{P}(u; g) = u - \int_0^T f(t) dt = 0$, where the data are $g = \{T, f(t)\}$.

A possible NP to approximate the integral in the MP is $\mathcal{P}_h(u_h; g_h) = u_h - h \sum_{i=0}^{N-1} f(t_i) = 0$, where $t_i = i h$ for

$i = 0, \dots, N$, with $h = \frac{T}{N}$ for some $N \in \mathbb{N}$. Here, $g_h = g$. We can also say that the size of the NP is N . The larger is N , the smaller is h , and the more accurate is the NP.

¹ If the NP is an iterative method, h is often replaced by the natural number n that refers to the number of iterations; in this case, we have $h \sim \frac{1}{n}$.

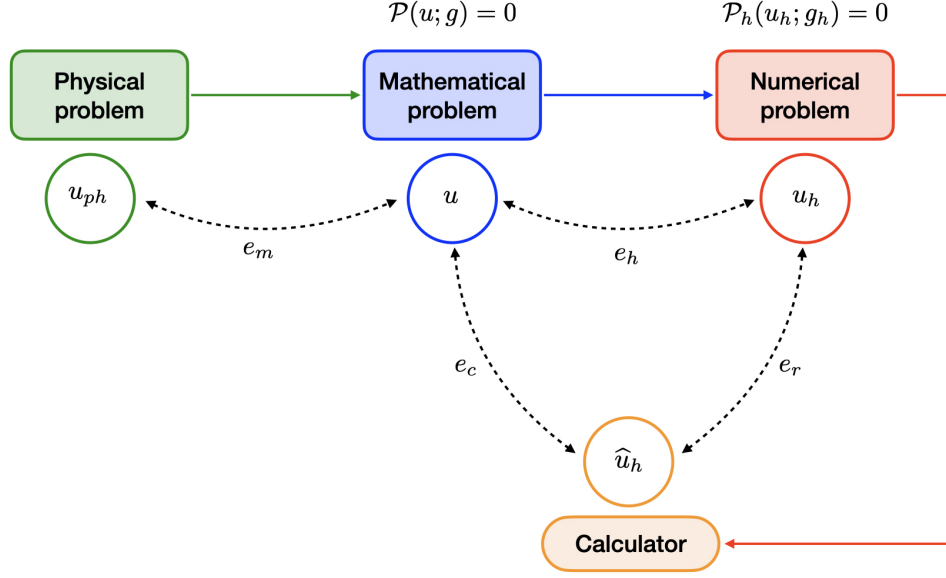


Figure 1.2: Physical (PP), mathematical (MP), and numerical (NP) problems. Corresponding solutions (u_{ph} , u , u_h , and $\hat{u}_h$) and errors (model $e_m = u_{ph} - u$, truncation $e_h = u - u_h$, round-off $e_r = u_h - \hat{u}_h$, and computational $e_c = e_h + e_r$ errors)

If the numerical solution is computed by executing the algorithm at the *calculator*, then the final solution is $\hat{u}_h$ and is affected by *round-off error*, say $e_r := u_h - \hat{u}_h$. Such round-off errors depend on the machine architecture, on the representation of the numbers at the calculator, and on operations made in floating-point arithmetic. Both the truncation and round-off errors concur to determine the *computational error*, say $e_c := u - \hat{u}_h = e_h + e_r$. For some NP, we can have $|e_r| \ll |e_h|$, for which $e_c \approx e_h$.

As for the MP, we have to ensure that the NP is well-posed and we have to assess the condition number of the NP.

Definition 1.2. The NP $\mathcal{P}_h(u_h; g_h) = 0$ of Eq. (1.9) is well-posed (stable) if and only if there exists a unique solution $u_h \in \mathcal{U}_h$ that continuously depends on the data $g_h \in \mathcal{G}_h$.

We are interested in NPs that allow to obtain computational errors that tend to zero as the numerical method “improves”, namely when the discretization parameter h goes to zero. This concept is encoded in the definition of *convergence*.

Definition 1.3. The NP is convergent when the computational error tends to zero for h tending to zero, that is:

$$\lim_{h \rightarrow 0} e_c = 0.$$

A crucial aspect is to qualify the convergence of the NP, that is determining the *convergence order* of the NP.

Definition 1.4. If $|e_c| \leq C h^p$, with C a positive constant independent of h and p , then the NP is convergent with order p .

If there exists a constant $\tilde{C} \leq C$ independent of h and p such that $\tilde{C} h^p \leq |e_c| \leq C h^p$, then we can write $|e_c| \simeq C h^p$ and we can estimate the convergence order p of the NP by using the known solution u of the MP. An approach to estimate p is *algebraic*. First, we compute the errors e_{c1} and e_{c2} for the NP

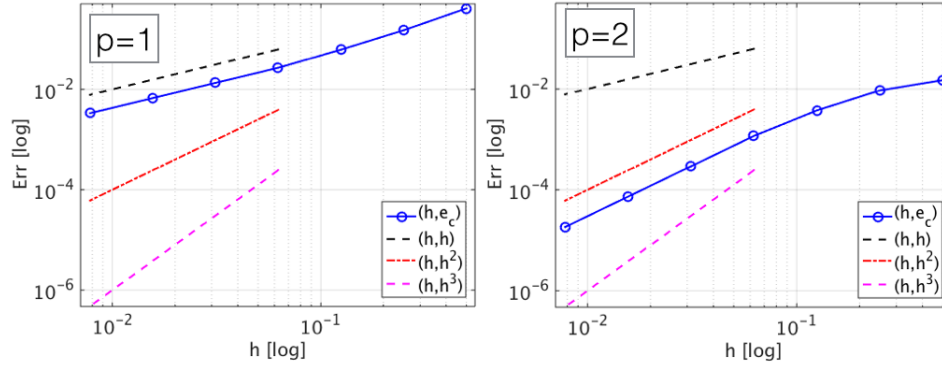


Figure 1.3: Graphical estimation of the convergence order p of a NP: computational errors $|e_c|$ vs. h

corresponding to two different values of h that are “sufficiently” small, say h_1 e h_2 , respectively; then, by writing $|e_{c1}| \simeq Ch_1^p$, $|e_{c2}| \simeq Ch_2^p$ and by noticing that $\frac{|e_{c1}|}{|e_{c2}|} = \left(\frac{h_1}{h_2}\right)^p$, we estimate the order p as:

$$p = \frac{\log(|e_{c1}|/|e_{c2}|)}{\log(h_1/h_2)}.$$

An alternative approach is based on the *graphical* estimate. We represent the errors $|e_c|$ vs. h on a plot in *log–log* scale. As $\log |e_c| = \log(Ch^p) = \log C + p \log h$, we have $p = \text{atan}(\theta)$, where θ is the slope of the curve (h, e_c) , a straight line in *log–log* scale. Instead of computing θ , it is possible to verify that the curves (h, e_c) and (h, h^p) are *parallel* in *log–log* scale; see Fig. 1.3.

We remark that a well-posed NP is not necessarily convergent. To ensure convergence of the NP, this is required to satisfy the *consistency* property: roughly speaking, the NP must be a “faithful copy” of the original MP.

Definition 1.5. The NP (1.9) is consistent if and only if $\lim_{h \rightarrow 0} \mathcal{P}_h(u; g) = \mathcal{P}(u; g) = 0$, with $g \in \mathcal{G}_h$.
The NP (1.9) is strongly consistent if and only if $\mathcal{P}_h(u; g) \equiv \mathcal{P}(u; g) = 0$ for all $h > 0$, with $g \in \mathcal{G}_h$.

It is clear that the NP must be well-posed (possibly, well-conditioned), consistent, and convergent. The concepts of well-posedness (stability) and convergence are indeed strongly connected and encoded in the following theoretical result.

Theorem 1.1 (Lax–Richtmeyer, equivalence). If the NP $\mathcal{P}_h(u_h; g_h) = 0$, with $u_h \in \mathcal{U}_h$ and $g_h \in \mathcal{G}_h$, is consistent, then it is well-posed if and only if it is also convergent.

It follows that, if the NP is well-posed and consistent, then the NP is also convergent. Equivalently, if the NP is consistent and convergent, then the NP is also well-posed. The equivalence theorem is very useful as it allows us to verify only two of the properties of a NP to obtain the third. In general, it is easier to show the consistency of a NP, while it is harder to show well-posedness and/or convergence.

The choice of a numerical method (PN) to approximate the solution u of a MP must take into account for: the mathematical properties of the MP; the computational efficiency in terms of expected convergence order of the error, the flops involved in the calculation, the performance of the CPU installed on the computer, the access modes and the availability of the calculation memory.

Let us indicate with N the size of the NP. Then, the number of floating-point operations required to calculate the numerical solution u_h depends on the size N of the NP. Different kinds of dependence are depicted in the following table.

	$O(1)$	$O(N)$	$O(N^\gamma)$	$O(\gamma^N)$	$O(N!)$
flops	independent	linear	polynomial	exponential	factorial

The following example highlights the role of selecting efficient numerical methods (NP) in relation with available computing resources.

Example 1.6. If the matrix $A \in \mathbb{R}^{N \times N}$ is non-singular, the solution $\mathbf{x} \in \mathbb{R}^N$ of the linear system $A\mathbf{x} = \mathbf{b}$, can be obtained by applying the *Cramer rule*:

$$x_i = \frac{\det(B_i)}{\det(A)} \quad i = 1, \dots, N,$$

where $B_i \in \mathbb{R}^{N \times N}$ is the matrix obtained from A by replacing its i -th column by the vector $\mathbf{b} \in \mathbb{R}^N$ as:

$$B_i = \begin{bmatrix} a_{11} & \dots & b_1 & \dots & a_{1n} \\ a_{21} & \dots & b_2 & \dots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & b_n & \dots & a_{nn} \end{bmatrix}.$$

$\uparrow$
 i

However, the solution of the linear system with this algorithm requires $O(e(N+1)!)$ floating-point operations. If $N = 100$, then about $101!e \approx 2.56 \cdot 10^{160}$ operations are required. A calculator able to perform 10^{12} floating-point operations (flops) per second – actually a supercomputer with a computing capability of 1 TeraFlop – will achieve the solution in $\frac{2.56 \cdot 10^{160}}{10^{12}} = 2.56 \cdot 10^{148}$ seconds, that is in about $8.11 \cdot 10^{140}$ years! More efficient numerical methods are available to solve the linear system, as for example the LU factorization method calls for $O\left(\frac{2}{3}N^3\right)$ flops. If $N = 100$, then the former supercalculator will provide the solution in only 10^{-6} seconds. A standard laptop will achieve the result in less than a second.

Chapter 2

Variational Formulation of Partial Differential Equations

In this chapter, we focus on the mathematical formulation of elliptic PDEs, in 1D or multidimensional domains (2D or 3D), with particular emphasis on the *variational formulation* of PDEs, also known as *weak formulation*. As we will see through some examples, the variational formulation is the proper setting to write a PDE and stands at the basis of the FE method. We will introduce some basic concepts of Functional Analysis to derive the weak formulation of the PDEs (starting from their strong formulation), to identify the appropriate function spaces (of infinite dimension) in which to look for the weak solutions, and to analyze such mathematical problem in terms of well-posedness. We will also consider some examples of weak formulations of elliptic PDEs, with different types of boundary conditions.

2.1 From the Strong Formulation to the Weak Formulation

Let us consider an elliptic (and stationary) PDE defined in a domain $\Omega \subset \mathbb{R}^d$, for $d = 1, 2$, or 3 , as discussed in Sec. 1.2. We can express such PDE as

$$\mathcal{L}u = g \quad \text{in } \Omega,$$

where $\mathcal{L}$ is a suitable differential operator and appropriate boundary conditions are set on the boundary $\partial\Omega$ of the domain Ω . We say that the former is the *strong formulation* of the PDE and that u is its *strong solution* if u satisfies the equation in *every* point $\mathbf{x} \in \Omega$ and every (partial) derivative appearing in $\mathcal{L}u$ is well-defined.

However, the strong formulation and the strong solution cannot represent many physically feasible solutions. In fact, a new formulation of the differential problem is needed, whose well-posedness analysis requires basic notions of function spaces. We show – through the following problems referring to the 1D elastic thread of Eq. (1.3) – that strong solutions of PDEs with physical meaning are not always admissible, unlike weak solutions. We also show that the basic calculus tool to derive the weak formulation from the strong one is integration by parts (partial integration).

2.1.1 Poisson problem in 1D: elastic thread

Let us consider again the Poisson problem (1.3) with homogeneous Dirichlet boundary conditions: find $u : \Omega \subset \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{cases} -u''(x) = f(x) & x \in \Omega = (a, b), \\ u(a) = u(b) = 0. \end{cases}$$

The former problem can represent the vertical displacement $u(x)$ in a point $x \in \overline{\Omega} = [a, b]$ of an elastic thread, loaded with the vertical force (per unit length) $f(x)$. If the function f in the former PDE is

“sufficiently” regular, for example if $f \in C^0(\Omega)$, then the strong solution is admitted and is such that $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$. Indeed, if $f \in C^0(\Omega)$, then $u'' \in C^0(\Omega)$ and $u \in C^2(\Omega)$. In this case, the strong formulation is well-defined and its strong solution is admitted.

However, if $f \notin C^0(\Omega)$, as for example in the case of concentrated loads or not uniformly distributed on the thread, the formulation (1.3) is no longer adequate. We remark that these situations typically represent realistic load configurations, with practical interest. We must therefore and in any case try to determine a solution u of the problem. This will be understood as *weak solution*, i.e. such that the relation $-u''(x) = f(x)$ must *not* be necessarily satisfied in every point $x \in \Omega$.

For the sake of simplicity, we will assume that the domain is $\Omega = (0, 1)$, i.e. $a = 0$ and $b = 1$.

Example: elastic thread with uneven, discontinuous load

We assume that the load f is a non uniform and discontinuous function defined as:

$$f(x) = \begin{cases} 0 & x \in \left(0, \frac{2}{5}\right) \text{ or } x \in \left[\frac{3}{5}, 1\right), \\ -1 & x \in \left[\frac{2}{5}, \frac{3}{5}\right). \end{cases}$$

Here, $f \notin C^0(0, 1)$, for which $u'' \notin C^0(0, 1)$. We have

$$u'(x) = \int_0^x u''(y) dy + C = - \int_0^x f(y) dy + C = -F(x) + C,$$

with $C \in \mathbb{R}$ a suitable constant. We obtain:

$$F(x) = \int_0^x f(y) dy = \begin{cases} 0 & x \in \left(0, \frac{2}{5}\right), \\ x - \frac{2}{5} & x \in \left[\frac{2}{5}, \frac{3}{5}\right), \\ \frac{1}{5} & x \in \left[\frac{3}{5}, 1\right), \end{cases}$$

and

$$u'(x) = \begin{cases} C & x \in \left(0, \frac{2}{5}\right), \\ x - \frac{2}{5} + C & x \in \left[\frac{2}{5}, \frac{3}{5}\right), \\ \frac{1}{5} + C & x \in \left[\frac{3}{5}, 1\right). \end{cases}$$

Thus, we have $u' \in C^0(0, 1)$. Similarly, $u(x) = \int_0^x u'(y) dy + \bar{C}$, where $\bar{C} \in \mathbb{R}$ is another suitable constant. Finally, we obtain:

$$u(x) = \begin{cases} Cx + D & x \in \left(0, \frac{2}{5}\right), \\ \frac{x^2}{2} - \frac{2}{5}x + Cx + E & x \in \left[\frac{2}{5}, \frac{3}{5}\right), \\ \frac{1}{5}x + Cx + F & x \in \left[\frac{3}{5}, 1\right), \end{cases}$$

with D, E , and $F \in \mathbb{R}$ constants playing the role of $\bar{C}$. As the elastic thread is continuous (we do not admit its rupture), we require that the vertical displacement is a continuous function of x , that is $u \in C^0(0, 1)$.

Moreover, we know from the boundary conditions that $u(0) = u(1) = 0$. We end up setting the following four conditions to determine the constants C , D , E , and F :

$$\left\{ \begin{array}{l} u(0) = 0, \\ u\left(\frac{2}{5}^{-}\right) = u\left(\frac{2}{5}^{+}\right), \\ u\left(\frac{3}{5}^{-}\right) = u\left(\frac{3}{5}^{+}\right), \\ u(1) = 0, \end{array} \right. \quad \text{that is} \quad \left\{ \begin{array}{l} D = 0, \\ \frac{2}{5}C = \frac{2}{25} - \frac{4}{25} + \frac{2}{5}C + E, \\ \frac{9}{50} - \frac{6}{25} + \frac{3}{5}C + E = \frac{3}{25} + \frac{3}{5}C + F, \\ \frac{1}{5} + C + F = 0. \end{array} \right.$$

We obtain that $C = -\frac{1}{10}$, $D = 0$, $E = \frac{2}{25}$, $F = -\frac{1}{10}$, and thus the solution reads:

$$u(x) = \begin{cases} -\frac{x}{10} & x \in \left[0, \frac{2}{5}\right) \\ \frac{x^2}{2} - \frac{x}{2} + \frac{2}{25} & x \in \left[\frac{2}{5}, \frac{3}{5}\right) \\ \frac{x}{10} - \frac{1}{10} & x \in \left[\frac{3}{5}, 1\right] \end{cases}$$

We verified that $u \in C^0([0, 1])$ and $u \in C^1(0, 1)$. However, u'' is not a continuous function in $\Omega = (0, 1)$, being undefined in $x = \frac{2}{5}$ and $\frac{3}{5}$. Thus, u is not a strong solution of the problem as $u \notin C^2(\Omega) \cap C^0(\overline{\Omega})$. Still, the solution u that we just determined has physical meaning and solves a problem with physical and practical interest (uneven, discontinuous load on an elastic thread).

Example: elastic thread with concentrated load

Let us assume that f is a concentrated load, that is a function corresponding to the *Dirac delta function*¹. Let us also assume that $f(x) = -\delta_0\left(x - \frac{1}{2}\right)$, that is f is a load of magnitude 1, downward directed, and centered in $x = 1/2$. We have:

$$u'(x) = \int_0^x u''(y) dy + C = - \int_0^x f(y) dy + C = -F(x) + C,$$

where

$$F(x) = \int_0^x f(y) dy = - \int_0^x \delta_0\left(y - \frac{1}{2}\right) dy = \begin{cases} 0 & x \in \left(0, \frac{1}{2}\right), \\ -1 & x \in \left[\frac{1}{2}, 1\right), \end{cases}$$

and then

$$u'(x) = \begin{cases} C & x \in \left(0, \frac{1}{2}\right), \\ C + 1 & x \in \left[\frac{1}{2}, 1\right). \end{cases}$$

¹The distribution *Dirac delta* centered in $x = 0$ is $\delta_0(x) = \lim_{n \rightarrow +\infty} g_n(x)$ where $g_n(x) = \begin{cases} 0 & |x| \geq 1/(2n), \\ n & |x| < 1/(2n), \end{cases}$ for $n \in \mathbb{N}$. This is such that: $\int_{-\infty}^{+\infty} g_n(x) dx = 1$ for all $n \in \mathbb{N}$, $\int_{-\infty}^{+\infty} \delta_0(x) dx = 1$, $\int_{-\infty}^{+\infty} h(x) \delta_0(x) dx = h(0)$, and $\int_{-\infty}^{+\infty} h(x) \delta_0(x - a) dx = h(a)$, for a given function $h(x)$.

Hence, $u' \notin C^0(0, 1)$. Moreover, from $u(x) = \int_0^x u'(y) dy + \bar{C}$, we obtain:

$$u(x) = \begin{cases} Cx + D & x \in \left(0, \frac{1}{2}\right), \\ (C+1)x + E & x \in \left[\frac{1}{2}, 1\right). \end{cases}$$

By setting the boundary conditions and the continuity of the elastic thread ($u \in C^0([0, 1])$), we have the following three conditions:

$$\begin{cases} u(0) = 0, \\ u\left(\frac{1}{2}^-\right) = u\left(\frac{1}{2}^+\right), \\ u(1) = 0, \end{cases} \quad \text{that is} \quad \begin{cases} D = 0, \\ \frac{C}{2} + D = \frac{C+1}{2} + E, \\ C+1+E = 0. \end{cases}$$

We obtain $C = E = -\frac{1}{2}$, and $D = 0$, from which

$$u(x) = \begin{cases} -\frac{x}{2} & x \in (0, 1/2), \\ \frac{x}{2} - \frac{1}{2} & x \in [1/2, 1). \end{cases} \quad (2.1)$$

Thus, $u \in C^0([0, 1])$, but $u \notin C^1(0, 1)$ nor $u \notin C^2(0, 1)$. The solution does not exist in the strong sense: u' and u'' are not defined in some points in the interval $(0, 1)$. However, the solution u of Eq. (2.1) is the *weak solution* of the problem.

Weak formulation

We need to introduce an alternative formulation of the problem in order to account for weak solutions.

Still by referring to the elastic thread problem – a Poisson problem with homogeneous boundary conditions – let us introduce a function $v \in C^1_0(0, 1)$ such that $v(0) = v(1) = 0$. We say that $v \in C^1_0(0, 1)$ and we call this function a *test function*. We multiply the differential problem by v and we integrate over $(0, 1)$, for which

$$\int_0^1 -u''(x) v(x) dx = \int_0^1 f(x) v(x) dx$$

must hold for every test function $v \in C^1_0(0, 1)$. By *partial integration* of the left hand side for a general 1D domain $\Omega = (a, b) \subset \mathbb{R}$, we have:

$$\begin{aligned} \int_a^b -u'' v dx &= \int_a^b -(u' v)' dx + \int_a^b u' v' dx = [-u' v]_{x=a}^b + \int_a^b u' v' dx \\ &= -[u'(b) v(b) - u'(a) v(a)] + \int_a^b u' v' dx. \end{aligned} \quad (2.2)$$

As $\Omega = (a, b) = (0, 1)$ and $v(0) = v(1) = 0$, we have in this case that:

$$\int_0^1 -u'' v dx = \int_0^1 u' v' dx.$$

This yields the *weak formulation* of the problem:

$$\text{find } u : \int_0^1 u' v' dx = \int_0^1 f v dx \quad \text{for all } v \in C^1_0(0, 1). \quad (2.3)$$

We will discuss and return later on the selection of the function spaces.

Remark 2.1. In the weak formulation of the problem, second derivatives of u do not appear. Therefore, it is no longer required that $u \in C^2(0, 1)$. In the weak formulation problem, u' and v' do appear instead, but under the integral: it is therefore not necessary that u' and v' are continuous functions.

Remark 2.2. The weak formulation (2.3) expresses the principle of least action, a variational principle in Mechanics that involves virtual works. When applied to the elastic thread, the solution consists in finding the displacement u such that the work of internal forces for any virtual displacement v , say $\int_0^1 u' v' dx$, is equal to the work of external forces done by the virtual displacement v , that is $\int_0^1 f v dx$.

Let us consider again the previous elastic thread problem with concentrated load. As $f(x) = -\delta_0\left(x - \frac{1}{2}\right)$, we have $\int_0^1 f(x) v(x) dx = -v\left(\frac{1}{2}\right)$ for all $v \in C_0^1(0, 1)$. From Eq. (2.1), we obtain:

$$u'(x) = \begin{cases} -1/2 & x \in (0, 1/2), \\ 1/2 & x \in [1/2, 1). \end{cases}$$

Hence, we have for all $v \in C_0^1(0, 1)$:

$$\begin{aligned} \int_0^1 u' v' dx &= \int_0^{1/2} u' v' dx + \int_{1/2}^1 u' v' dx = -\frac{1}{2} \int_0^{1/2} v' dx + \frac{1}{2} \int_{1/2}^1 v' dx \\ &= -\frac{1}{2} \left[v\left(\frac{1}{2}\right) - v(0) \right] + \frac{1}{2} \left[v(1) - v\left(\frac{1}{2}\right) \right] = -v\left(\frac{1}{2}\right), \end{aligned}$$

as $v(1) = v(0) = 0$. Thus, we just verified that the solution (2.1) is a weak solution of the weak formulation of the elastic thread problem with concentrated load.

2.1.2 Advection–diffusion–reaction problem in 1D

Let us consider the following 1D advection–diffusion–reaction problem, with homogeneous Dirichlet boundary conditions and non uniform coefficients:

$$\begin{cases} -(\mu(x) u'(x))' + \beta(x) u'(x) + \sigma(x) u(x) = f(x) & x \in \Omega = (a, b) \subset \mathbb{R}, \\ u(a) = u(b) = 0. \end{cases}$$

Here, $\mu(x) \geq \mu_0 > 0$ and $\sigma(x) \geq \sigma_0 > 0$ for every $x \in (a, b)$. The terms at the left hand side represent diffusion, advection, and reaction of a field u , respectively. If $\beta = \sigma = 0$, we find a model similar to that of the elastic thread, even if with elastic properties encoded in the function $\mu(x)$, possibly varying along the thread.

If μ', β, σ , and $f \in C^0(\Omega)$, then the former strong formulation is well-defined and a strong solution $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$ is admitted. However, data of practical interest are often not “sufficiently” regular to guarantee the existence of a strong solution. This occurs for example if $\mu(x)$ is discontinuous. In this case, a weak solution may be determined by rewriting the problem in weak formulation.

As seen for the elastic thread problem, in order to determine the weak formulation, we ask that:

$$\int_a^b [-(\mu u')' + \beta u' + \sigma u] v dx = \int_a^b f v dx \quad \text{for all } v \in C_0^1(\Omega).$$

Following Eq. (2.2), by partial integration of the first term, and by recalling that $v(a) = v(b) = 0$, we obtain the weak formulation:

$$\text{find } u : \int_a^b \mu u' v' dx + \int_a^b \beta u' v dx + \int_a^b \sigma u v dx = \int_a^b f v dx \quad \text{for all } v \in C_0^1(\Omega). \quad (2.4)$$

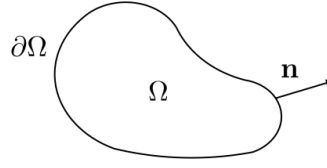


Figure 2.1: A domain $\Omega \subset \mathbb{R}^2$, its boundary $\partial\Omega$, and the outward directed, unit vector $\mathbf{n}$ normal to the boundary $\partial\Omega$

Once again, we recall that, in the former weak formulation, u'' does not appear and u' , v' only appear under integrals: u' and v' are not required to be continuous functions. Similarly, it is not required that μ' , β , σ , and f are continuous functions to yield a well-defined weak formulation. In summary, in the weak formulation, requirements of regularity of the data are much less stringent than those of the strong formulation.

Remark 2.3. *If there exists a strong solution of the problem, i.e. $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$, then this is also a weak solution (i.e. it also satisfies the problem in weak formulation). Instead, the weak solution of a problem is not a strong solution, in general.*

2.1.3 Poisson equation in 2D: elastic membrane

Let us consider again the Poisson problem (1.5) with homogeneous Dirichlet boundary conditions: find $u : \Omega \subset \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega; \end{cases}$$

here, $d = 2$ or 3 . The domain $\Omega \subset \mathbb{R}^d$ is endowed with boundary $\partial\Omega$. We indicate by $\mathbf{n}$ the outward directed, unit vector normal to the boundary $\partial\Omega$; see for example Fig. 2.1. If $d = 2$, the model represents an elastic membrane, fixed at the boundaries, which undergoes a vertical displacement u under the action of an external load f (force per unit area).

We write the weak formulation of the former problem. First, we introduce the test function $v \in C_0^1(\Omega)$, i.e. such that ∇v is a continuous vector-valued function in Ω and $v = 0$ on $\partial\Omega$. Then, we multiply the PDE by v and we integrate, thus obtaining:

$$\int_{\Omega} -\Delta u(\mathbf{x}) v(\mathbf{x}) d\mathbf{x} = \int_{\Omega} f(\mathbf{x}) v(\mathbf{x}) d\mathbf{x}.$$

Partial integration of the left hand side yields the following formula:

$$\begin{aligned} \int_{\Omega} -\Delta u v d\mathbf{x} &= \int_{\Omega} -\nabla \cdot (v \nabla u) d\mathbf{x} + \int_{\Omega} \nabla u \cdot \nabla v d\mathbf{x} \\ &= \oint_{\partial\Omega} -v \nabla u \cdot \mathbf{n} d\mathbf{x} + \int_{\Omega} \nabla u \cdot \nabla v d\mathbf{x}. \end{aligned} \quad (2.5)$$

The term $\nabla u \cdot \mathbf{n}$ on the boundary $\partial\Omega$ is often indicated as $\frac{\partial u}{\partial n}$.

By recalling that $v = 0$ on $\partial\Omega$, we obtain the weak formulation of the Poisson problem:

$$\text{find } u : \int_{\Omega} \nabla u \cdot \nabla v d\mathbf{x} = \int_{\Omega} f v d\mathbf{x} \quad \text{for all } v \in C_0^1(\Omega). \quad (2.6)$$

We still have to identify the function space to which the weak solution u belongs.

2.2 Basic Notions of Functional Analysis

In the previous section, we derived the weak formulations of some elliptic PDEs. We now have to mathematically justify the procedure, to establish under which hypotheses the formulation of the problem is mathematically well-posed, to choose the appropriate function spaces for the weak solution and test function, and finally give sense to the notions of integrals and derivatives. The weak formulation of a PDE involves a significant change of perspective, for which it is necessary to introduce some new concepts of Functional Analysis. We will conclude the section by proving the Lax–Milgram theorem, which provides the tools for the well-posedness analysis of the weak formulation.

The first concept to be introduced is the *Lebesgue integral*, which extends the class of functions that can be integrated (with respect to the standard Riemann integral). The fundamental difference between the Lebesgue integral and the Riemann one consists in the choice of decompositions on which each integral is based: the Riemann integral decomposes the integration interval into subintervals, while the Lebesgue integral decomposes into subintervals the image of the integrated function. According to the Lebesgue integral, for a domain $\Omega \subset \mathbb{R}^d$, if f and $g : \Omega \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ differ on a set of zero measure, then $f = g$ almost everywhere (a.e.) in Ω and $\int_{\Omega} f \, dx = \int_{\Omega} g \, dx$. Moreover, f can be integrated if and only if $|f|$ can be integrated. If f admits improper integral according to Riemann, then f can be integrated according to Lebesgue.

We will implicitly assume that integrals in the weak formulation are Lebesgue integrals. Functions $f : \Omega \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$, which admit Lebesgue integrals, belong to a function space indicated as $L^1(\Omega)$.

2.2.1 Function spaces

A *function space* is a set of functions between two fixed sets. It is a particular (linear) *vector space*, which is a set whose elements, called vectors, can be added together and multiplied by numbers (scalars). Our goal is to introduce a framework to define those functions – the weak solution and test function – that appear in the weak formulation of the PDEs.

Banach spaces

Let us introduce the notion of Banach space.

Definition 2.1. Given a vector space X on $\mathbb{R}$, the norm over X is a function $\|\cdot\| : X \rightarrow \mathbb{R}$ such that, for all $\lambda \in \mathbb{R}$ and for all $x, y \in X$, the following properties hold:

$$\begin{aligned} N_1 : \|x\| &\geq 0 \quad \text{and} \quad \|x\| = 0 \text{ if and only if } x = 0 && (\text{positivity}); \\ N_2 : \|\lambda x\| &= |\lambda| \|x\| && (\text{homogeneity}); \\ N_3 : \|x + y\| &\leq \|x\| + \|y\| && (\text{triangular inequality}). \end{aligned}$$

A normed vector space X is a vector space endowed with a norm $\|\cdot\|$.

A seminorm of the vector space X is a function $|\cdot| : X \rightarrow \mathbb{R}$ satisfying the properties N_2 , N_3 , and the following one: $|x| \geq 0$ for all $x \in X$ (in place of N_1).

The following definition introduces the notion of equivalence of two norms of the space X .

Definition 2.2. Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ of the space X are equivalent if there exist two constants c_1 and $c_2 > 0$ such that

$$c_1 \|x\|_2 \leq \|x\|_1 \leq c_2 \|x\|_2 \quad \text{for all } x \in X.$$

The vector space X is a *metric space* if endowed with the *distance* function between each of its elements $x, y \in X$, i.e. there exists the distance induced by the norm $\|\cdot\|$ of the space X , say $d(x, y) = \|x - y\|$.

An important aspect is the *completeness* of the space X . In particular, we will address this property for the spaces $C^k(\Omega)$, that is spaces of functions $f : \Omega \rightarrow \mathbb{R}$ that are k -times continuously differentiable.

Definition 2.3. A sequence $\{x_n\}_{n \geq 1}$ of elements in the normed space X is convergent to an element $x \in X$, that is $\lim_{n \rightarrow +\infty} x_n = x \in X$ if and only if

$$\|x_n - x\| \rightarrow 0 \quad \text{for } n \rightarrow +\infty.$$

A sequence $\{x_n\}_{n \geq 1}$ of elements in the normed space X is called Cauchy sequence (fundamental sequence) if and only if

$$\|x_m - x_n\| \rightarrow 0 \quad \text{for } m, n \rightarrow +\infty.$$

Definition 2.4. The normed space X is complete if all the Cauchy sequences $\{x_n\}_{n \geq 1}$ in the space X are convergent.

Definition 2.5. A linear vector space X that is normed and complete is called Banach space.

Among the Banach spaces, we consider the following ones for functions $f : \Omega \subset \mathbb{R}^d \rightarrow \mathbb{R}$, for $d = 1, 2$, and 3 .

Definition 2.6. The space $L^p(\Omega)$ of functions $f : \Omega \rightarrow \mathbb{R}$ such that

$$\int_{\Omega} |f|^p d\mathbf{x} < +\infty$$

is a Banach space with norm

$$\|f\|_{L^p(\Omega)} := \left(\int_{\Omega} |f|^p d\mathbf{x} \right)^{1/p} \quad \text{for } p = 1, 2, \dots, +\infty.$$

In particular, we have already seen that $L^1(\Omega)$ is the space of functions that are integrable according to Lebesgue with norm $\|f\|_{L^1(\Omega)} = \int_{\Omega} |f| d\mathbf{x}$. The space $L^2(\Omega)$ is endowed with the norm:

$$\|f\|_{L^2(\Omega)} = \sqrt{\int_{\Omega} |f|^2 d\mathbf{x}}.$$

Moreover, $L^\infty(\Omega)$ is the space of measurable functions bounded almost everywhere, i.e. such that $|f(\mathbf{x})| \leq M < +\infty$ a.e. in Ω , with norm:

$$\|f\|_{L^\infty(\Omega)} = \sup_{\mathbf{x} \in \Omega} \{|f(\mathbf{x})| \text{ a.e. in } \Omega\}.$$

Remark 2.4. The spaces $L^p(\Omega)$ are such that $L^q(\Omega) \subseteq L^p(\Omega) \subseteq L^1(\Omega)$ for $1 \leq p \leq q \leq +\infty$.

Hilbert spaces

Hilbert spaces are Banach spaces with a richer structure that represent the natural framework for the weak formulation of PDEs. The rich structure of Hilbert spaces comes from the fact that these are endowed with the notions of scalar product and an induced norm. The mathematical concept of a Hilbert space generalizes the notion of Euclidean space $\mathbb{R}^d$, and allows to extend the methods of algebra and vector

calculus to both finite and infinite dimensional spaces. Indeed, a Hilbert space is an abstract vector space that possesses the structure of a scalar (or inner) product that allows lengths and angles to be measured.

As a matter of fact, the function space $C_0^1(\Omega)$ that we initially selected for the test functions in the weak formulations (2.3), (2.4), and (2.6) is not a Hilbert space and is not the appropriate functional setting for such mathematical problem.

Definition 2.7. Given a linear vector space X over $\mathbb{R}$, a scalar product (inner product) on X is a function $(\cdot, \cdot) : X \times X \rightarrow \mathbb{R}$ such that, for all λ and $\mu \in \mathbb{R}$ and for all $x, y, z \in X$, the following properties hold

$$I_1 : (x, x) \geq 0 \text{ and } (x, x) = 0 \text{ if and only if } x = 0 \quad (\text{positivity});$$

$$I_2 : (x, y) = (y, x) \quad (\text{symmetry});$$

$$I_3 : (\lambda x + \mu y, z) = \lambda (x, z) + \mu (y, z) \quad (\text{bilinearity}).$$

The scalar product induces a norm for the vector space X , that is:

$$\|x\| = \sqrt{(x, x)} \quad \text{for all } x \in X.$$

Definition 2.8. If X is a vector space endowed with scalar product, then X is called Hilbert space if it is complete with respect to the norm induced by the scalar product.

Theorem 2.1 (Cauchy–Schwarz inequality). For any Hilbert space X , the following inequality holds:

$$|(x, y)| \leq \|x\| \|y\| \quad \text{for all } x, y \in X. \quad (2.7)$$

Example 2.1. The space $X = \mathbb{R}^d$ with Euclidean norm $\|\cdot\|_2$ is a Hilbert space, i.e.: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^d x_i^2} = \sqrt{(\mathbf{x}, \mathbf{x})}$ for all $\mathbf{x} = (x_i)_{i=1}^d \in \mathbb{R}^d$, where $(\cdot, \cdot)$ is the scalar product among vectors in $\mathbb{R}^d$.

Given an open and bounded set $\Omega \subset \mathbb{R}^d$, with $d = 1, 2$, or 3 , the Banach space $L^2(\Omega)$ with norm $\|\cdot\|_{L^2(\Omega)}$ is also a Hilbert space. Its scalar product is

$$(f, g)_{L^2(\Omega)} = \int_{\Omega} f g \, d\mathbf{x} \quad \text{for all } f, g \in L^2(\Omega),$$

while the norm induced by the scalar product is:

$$\|f\|_{L^2(\Omega)} = \sqrt{\int_{\Omega} f^2 \, d\mathbf{x}} = \sqrt{(f, f)_{L^2(\Omega)}} \quad \text{for all } f \in L^2(\Omega).$$

Remark 2.5. The Banach spaces $L^p(\Omega)$ for $p \neq 2$ are not Hilbert spaces. The spaces $C^0(\overline{\Omega})$ and $C_0^1(\Omega)$ are not Hilbert spaces.

Sobolev spaces

Sobolev spaces are subsets of Hilbert spaces that represent the ideal framework for setting the weak formulation of PDEs.

Definition 2.9. Given $\Omega \subset \mathbb{R}^d$, with $d \geq 1$, and the function $v : \Omega \rightarrow \mathbb{R}$, then the Sobolev space $H^1(\Omega)$ is defined as

$$H^1(\Omega) := \{v \in L^2(\Omega) : \nabla v \in [L^2(\Omega)]^d\};$$

if $d = 1$ ($\Omega \subset \mathbb{R}$), we have:

$$H^1(\Omega) := \{v \in L^2(\Omega) : v' \in L^2(\Omega)\}.$$

We remark that, for $d > 1$, ∇v is a vector-valued function with d components: asking that $\nabla v \in [L^2(\Omega)]^d$ means that each component of ∇v belongs to $L^2(\Omega)$.

Definition 2.10. Given $\Omega \subset \mathbb{R}^d$, with $d \geq 1$, and the function $v : \Omega \rightarrow \mathbb{R}$, the seminorm $H^1(\Omega)$ is:

$$|v|_{H^1(\Omega)} := \|\nabla v\|_{L^2(\Omega)} = \left(\int_{\Omega} |\nabla v|^2 d\mathbf{x} \right)^{1/2} \quad \text{if } d > 1,$$

$$|v|_{H^1(\Omega)} := \|v'\|_{L^2(\Omega)} = \left(\int_{\Omega} (v')^2 dx \right)^{1/2} \quad \text{if } d = 1.$$

Definition 2.11. Given $\Omega \subset \mathbb{R}^d$, with $d \geq 1$, and the functions $w, v : \Omega \rightarrow \mathbb{R}$, the Hilbert space $H^1(\Omega)$ is endowed with the following scalar product:

$$(w, v)_{H^1(\Omega)} := \int_{\Omega} w v d\mathbf{x} + \int_{\Omega} \nabla w \cdot \nabla v d\mathbf{x},$$

and the norm induced by the scalar product is:

$$\|v\|_{H^1(\Omega)} = \sqrt{\|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2} = \sqrt{\int_{\Omega} v^2 d\mathbf{x} + \int_{\Omega} |\nabla v|^2 d\mathbf{x}}.$$

If $d = 1$, then

$$(w, v)_{H^1(\Omega)} = \int_{\Omega} w v dx + \int_{\Omega} w' v' dx$$

and

$$\|v\|_{H^1(\Omega)} = \sqrt{\|v\|_{L^2(\Omega)}^2 + \|v'\|_{L^2(\Omega)}^2} = \sqrt{\int_{\Omega} v^2 dx + \int_{\Omega} (v')^2 dx}.$$

More in general, we give the following definition of Sobolev spaces $H^k(\Omega)$.

Definition 2.12. Given $\Omega \subset \mathbb{R}^d$, with $d \geq 1$, then the Sobolev space $H^k(\Omega)$ is defined as:

$$H^k(\Omega) := \{v \in L^2(\Omega) : D^{\alpha} v \in L^2(\Omega) \text{ for all } \alpha : |\alpha| \leq k\};$$

if $d = 1$ ($\Omega \subset \mathbb{R}$), we have:

$$H^k(\Omega) = \{v \in L^2(\Omega) : v^{(\alpha)} \in L^2(\Omega) \text{ for all } \alpha = 0, \dots, k\}.$$

For $d > 1$, $\alpha = (\alpha_1, \dots, \alpha_d)$ is a multi-index, where α_i is a non negative integer, such that $|\alpha| = \alpha_1 + \dots + \alpha_d$; the components of α correspond to the order of derivation.

To give sense to the definitions of Sobolev spaces, we need to introduce a new notion of *derivative* of a function. Derivatives are indeed to be intended as *weak derivatives*, or derivatives in the sense of *distributions*. For simplicity, let us consider the case of a function $f : \Omega \rightarrow \mathbb{R}$ for which $\Omega = (a, b) \subset \mathbb{R}$. Then, the weak derivative of v is a function w such that

$$\int_a^b w \varphi dx = - \int_a^b v \varphi' dx \quad \text{for all } \varphi \in C_0^\infty(\Omega),$$

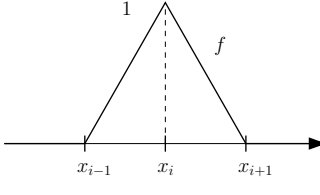
where $C_0^\infty(\Omega)$ is the set of functions infinitely differentiable with zero value at the boundaries². If w exists, then it is the weak derivative of v and we can write $w = v'$ in the sense of distributions. If $v \in C^1(\Omega)$, then the classic derivative coincides with the weak derivative. Conversely, if $v \notin C^1(\Omega)$, v may admit a weak derivative even if it does not admit a classic derivative.

Example 2.2. The weak derivative of the Heaviside step function $H(x)$, a discontinuous function, is the Dirac delta function centered in zero, i.e. $w(x) = H'(x) = \delta_0(x)$. Indeed, for $\Omega = (a, b)$ such that $0 \in \Omega$, we have:

$$\int_a^b w \varphi dx = - \int_a^b H \varphi' dx = - \int_0^b \varphi' dx = \varphi(0) - \varphi(b) = \varphi(0) = \int_{\mathbb{R}} \delta_0 \varphi dx \quad \text{for all } \varphi \in C_0^\infty(\Omega).$$

We remark that the classic derivative of $H(x)$ does not exist, while its weak derivative is well-defined.

Example 2.3. Let us consider three nodes x_{i-1} , x_i , and x_{i+1} such that $x_i - x_{i-1} = x_{i+1} - x_i = h$ and the function $f \in L^2(\mathbb{R})$ defined as:

$$f(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}} & \text{if } x_{i-1} \leq x \leq x_i, \\ \frac{x_{i+1} - x}{x_{i+1} - x_i} & \text{if } x_i \leq x \leq x_{i+1}, \\ 0 & \text{otherwise.} \end{cases}$$


The function f is continuous, but $f \notin C^1(\mathbb{R})$. In weak sense, we can compute the derivative as:

$$\begin{aligned} \int_{\mathbb{R}} w \varphi dx &= - \int_{\mathbb{R}} f \varphi' dx = - \int_{x_{i-1}}^{x_{i+1}} f \varphi' dx = - \int_{x_{i-1}}^{x_i} f \varphi' dx - \int_{x_i}^{x_{i+1}} f \varphi' dx \\ &= \int_{x_{i-1}}^{x_i} f' \varphi dx + \int_{x_i}^{x_{i+1}} f' \varphi dx = \frac{1}{h} \left(\int_{x_{i-1}}^{x_i} \varphi dx - \int_{x_i}^{x_{i+1}} \varphi dx \right), \end{aligned}$$

from which $w = \frac{1}{h} (\chi_{(x_{i-1}, x_i]} - \chi_{(x_i, x_{i+1}]})$, where the characteristic function is $\chi_{[a, b]}(x) = 1$ if $x \in (a, b]$ and zero otherwise. The former w is the weak derivative of f , say f' . Actually, $f \in H^1(\mathbb{R})$.

The notion of weak derivative can be extended to functions in $\Omega \subset \mathbb{R}^d$, with $d \geq 1$. In general, from now on, derivatives of functions will be intended in the weak sense.

For $\Omega = (a, b) \subset \mathbb{R}$, i.e. $d = 1$, the functions $v \in H^1(\Omega)$ are also continuous. Therefore, $v(x_0)$ is well-defined in every point $x_0 \in \overline{\Omega} = [a, b]$. This is no longer true for $\Omega \subset \mathbb{R}^d$, with $d \geq 2$. In general, for $d \geq 1$, we have the following Sobolev immersion result that justifies the former statements.

Proposition 2.1. Let $\Omega \subset \mathbb{R}^d$, with $d \geq 1$, be an open and bounded domain endowed with a “sufficiently” regular border $\partial\Omega$. Then,

$$H^k(\Omega) \subset C^m(\overline{\Omega}) \quad \text{if } k > m + \frac{d}{2}.$$

²The weak derivative w of a function v leverages on the partial integration formula (2.2) and the choice $\varphi \in C_0^\infty(\Omega)$ such that $\varphi(a) = \varphi(b) = 0$.

Remark 2.6. In the former, let us set $m = 0$, then $H^k(\Omega) \subset C^0(\overline{\Omega})$ if $k > \frac{d}{2}$. We have, for $d = 1$, that functions in $H^1(\Omega)$ are continuous. Conversely, for $d = 2$ and 3 , there exist functions in $H^1(\Omega)$ that are not continuous.

Let us define a subspace of $H^1(\Omega)$ over an open domain $\Omega \subset \mathbb{R}^d$, with $d \geq 1$. The following definition will be important to define function spaces for PDEs with Dirichlet boundary conditions.

Definition 2.13. We indicate by $H_0^1(\Omega)$ the space

$$H_0^1(\Omega) := \{v \in H^1(\Omega) : v = 0 \text{ on } \partial\Omega\};$$

otherwise, for $d = 1$:

$$H_0^1(a, b) = \{v \in H^1(a, b) : v(a) = v(b) = 0\}.$$

We notice that, for $d \geq 2$, the functions $v \in H^1(\Omega)$ are not necessarily continuous, and therefore their value on $\partial\Omega$ may not be well-defined in the classic sense. However, it is possible to define a suitable *trace operator* – or simply *trace* – that associates to every function $v \in H^1(\Omega)$ its restriction γv on $\partial\Omega$ in a unique and well-defined manner. Instead, this is not a problem for $d = 1$, since in that case functions in $H^1(\Omega)$ are also continuous; indeed, for $d = 1$, we have $v \in C^0(\overline{\Omega})$ and $\gamma v \equiv v$ on $\partial\Omega$. The following trace theorem holds.

Theorem 2.2 (trace inequality). If $\Omega = (a, b) \subset \mathbb{R}$, then there exists a positive constant $\tilde{C}_T > 0$, such that

$$|v(a)| \leq \tilde{C}_T \|v\|_{H^1(a, b)} \quad \text{and} \quad |v(b)| \leq \tilde{C}_T \|v\|_{H^1(a, b)} \quad \text{for all } v \in H^1(a, b). \quad (2.8)$$

For $\Omega \subset \mathbb{R}^d$, with $d \geq 2$, we have:

$$\|\gamma v\|_{L^2(\partial\Omega)} \leq \tilde{C}_T \|v\|_{H^1(\Omega)} \quad \text{for all } v \in H^1(\Omega),$$

where γv indicates the trace of v on $\partial\Omega$.

A significant property of the subspace $H_0^1(\Omega)$ is encoded in the *Poincaré inequality*.

Theorem 2.3 (Poincaré inequality). For an open and bounded domain $\Omega \subset \mathbb{R}^d$, with $d \geq 1$, there exists a positive constant $C_\Omega > 0$ – called *Poincaré constant* – such that:

$$\|v\|_{L^2(\Omega)} \leq C_\Omega \|v\|_{H^1(\Omega)} \quad \text{for all } v \in H_0^1(\Omega). \quad (2.9)$$

Remark 2.7. The Poincaré inequality also holds for $v \in H_{\Gamma_D}^1(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}$, where $\Gamma_D \subset \partial\Omega$ is a subset of the border $\partial\Omega$. In this case, there exists a constant $C_\Omega > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C_\Omega \|v\|_{H^1(\Omega)} \quad \text{for all } v \in H_{\Gamma_D}^1(\Omega).$$

The following lemma follows from the Poincaré inequality.

Lemma 2.1. The seminorm $|\cdot|_{H^1(\Omega)}$ is a norm of the space $H_0^1(\Omega)$ (or $H_{\Gamma_D}^1(\Omega)$) since it is equivalent to the norm $\|\cdot\|_{H^1(\Omega)}$ as for Definition 2.2.

Proof. Let us consider for simplicity the case $d = 1$. We have:

$$\|v\|_{H^1(\Omega)} = \sqrt{\|v\|_{L^2(\Omega)}^2 + \|v'\|_{L^2(\Omega)}^2} = \sqrt{\|v\|_{L^2(\Omega)}^2 + |v|_{H^1(\Omega)}^2} \quad \text{for all } v \in H_0^1(\Omega).$$

Then, by using the H^1 norm definition and the Poincaré inequality, we have

$$\|v\|_{H^1(\Omega)}^2 = \|v\|_{L^2(\Omega)}^2 + |v|_{H^1(\Omega)}^2 \leq (1 + C_\Omega^2) |v|_{H^1(\Omega)}^2.$$

Finally, we obtain the equivalence of the norm and seminorm:

$$|v|_{H^1(\Omega)} \leq \|v\|_{H^1(\Omega)} \leq \sqrt{1 + C_\Omega^2} |v|_{H^1(\Omega)} \quad \text{for all } v \in H_0^1(\Omega).$$

□

2.2.2 Linear functionals and bilinear forms

Functionals and forms are essential objects in the weak formulation of a PDE. These operate on Hilbert or Sobolev spaces that, in this section, we will indicate as V . The corresponding norm, induced by the scalar product, will be indicated as $\|\cdot\|_V$.

Let us start with the notion of linear functional $F(\cdot)$ over the Hilbert space V .

Definition 2.14. Given the Hilbert space V , a linear functional is an operator $F : V \rightarrow \mathbb{R}$ such that

$$F(\beta v + \gamma w) = \beta F(v) + \gamma F(w) \quad \text{for all } \beta, \gamma \in \mathbb{R} \quad \text{and for all } v, w \in V.$$

Example 2.4. In the weak formulation (2.3) of the 1D Poisson problem, the right hand side is a linear functional $F(v) = \int_0^1 f v \, dx$, if $v \in V$, with V a Hilbert space.

Definition 2.15. Given a linear functional $F : V \rightarrow \mathbb{R}$, we say that

- F is bounded if and only if there exists a constant $C > 0$ such that:

$$|F(v)| \leq C \|v\|_V \quad \text{for all } v \in V;$$

- F is continuous if and only if there exists a constant $\tilde{C} > 0$ such that

$$|F(v) - F(w)| \leq \tilde{C} \|v - w\|_V \quad \text{for all } v, w \in V.$$

Proposition 2.2. The linear functional $F : V \rightarrow \mathbb{R}$ is continuous if and only if it is also bounded.

Definition 2.16. Given a Hilbert space V , its dual space V' is the set of all linear and continuous functionals $F : V \rightarrow \mathbb{R}$. The dual space V' is endowed with the following norm:

$$\|F\|_{V'} = \sup_{v \in V \setminus \{0\}} \frac{|F(v)|}{\|v\|_V}.$$

Moreover, the following result allows identifying the Hilbert space V with its dual space V' .

Theorem 2.4 (Riesz representation). Given the Hilbert space V , for all $F \in V'$, there exists a unique element (a function) $f \in V$ such that $F(v) = (f, v)_V$ for all $v \in V$.

Let us introduce the bilinear form $a(\cdot, \cdot)$ over the Hilbert space V .

Definition 2.17. Given the Hilbert space V , a bilinear form over V is an application $a : V \times V \rightarrow \mathbb{R}$ such that

$$a(\beta v + \gamma w, z) = \beta a(v, z) + \gamma a(w, z) \quad \text{and} \quad a(v, \beta w + \gamma z) = \beta a(v, w) + \gamma a(v, z),$$

for all $\beta, \gamma \in \mathbb{R}$ and for all $v, w, z \in V$.

Example 2.5. In the weak formulation (2.3) of the 1D Poisson problem, the left hand side represents a bilinear form $a(u, v) = \int_0^1 u' v' dx$, if $u, v \in V$, a Hilbert space.

Definition 2.18. Given a bilinear form $a : V \times V \rightarrow \mathbb{R}$, we say that:

- a is symmetric if and only if $a(v, w) = a(w, v)$ for all $v, w \in V$;
- a is continuous if and only if there exists a continuity constant $M > 0$ such that

$$|a(v, w)| \leq M \|v\|_V \|w\|_V \quad \text{for all } v, w \in V;$$

- a is coercive if and only if there exists a positive coercivity constant $\alpha > 0$ such that

$$a(v, v) \geq \alpha \|v\|_V^2 \quad \text{for all } v \in V.$$

Example 2.6. Let us consider the space $V = H^1(\Omega)$, with $\Omega \subset \mathbb{R}$. The bilinear form $a(u, v) = \int_{\Omega} u' v' dx + \int_{\Omega} u v dx$ is symmetric. The bilinear form $a(u, v) = \int_{\Omega} u' v dx$ is not symmetric.

2.2.3 Lax–Milgram theorem

We will see that the weak formulation of a linear, elliptic PDE is well-defined if the weak solution u and the test function v lay both in a suitable Hilbert (Sobolev) space V .

The weak formulation of an elliptic PDE generally reads:

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V, \quad (2.10)$$

where $a : V \times V \rightarrow \mathbb{R}$ is a bilinear form and $F : V \rightarrow \mathbb{R}$ is a linear functional.

Under some hypothesis on V , $a(\cdot, \cdot)$, and $F(\cdot)$, the well-posedness of the weak problem is guaranteed by the following result.

Theorem 2.5 (Lax–Milgram theorem). Given the Hilbert space V with norm $\|\cdot\|_V$ induced by the scalar product $(\cdot, \cdot)$, if all the following hypothesis hold

- i) $a : V \times V \rightarrow \mathbb{R}$ is a bilinear form;
- ii) $a : V \times V \rightarrow \mathbb{R}$ is continuous;
- iii) $a : V \times V \rightarrow \mathbb{R}$ is coercive;
- iv) $F : V \rightarrow \mathbb{R}$ is a linear functional;
- v) $F : V \rightarrow \mathbb{R}$ is continuous;

then there exists a unique weak solution $u \in V$ of the weak problem (2.10) and the following stability estimate holds:

$$\|u\|_V \leq \frac{1}{\alpha} \|F\|_{V'}. \quad (2.11)$$

The Lax–Milgram theorem stands as the theoretical tool to show the existence and uniqueness of the weak solution $u \in V$ of elliptic PDEs in weak formulation.

2.3 Weak Formulation of Elliptic Partial Differential Equations

We have introduced in the previous section basic notions of functional analysis that allow to rewrite PDEs in weak formulation and to enable their well-posedness analysis in virtue of the Lax–Milgram theorem. We consider elliptic PDEs in space domains $\Omega \subset \mathbb{R}^d$, with $d = 1, 2$, and 3 . In particular, we consider different types of boundary conditions: Dirichlet, Neumann, and Robin and their mixed forms.

2.3.1 PDEs in 1D

We consider different examples of elliptic PDEs in 1D, for which for example $\Omega = (0, L) \subset \mathbb{R}$. This family of differential models is also known as boundary value problems in 1D. For some of these problems, we will provide a detailed well-posedness analysis.

Poisson problem with homogeneous Dirichlet boundary conditions

Let us generalize the Poisson problem of (1.3) and Sec. 2.1.1, still with homogeneous Dirichlet boundary conditions:

$$\begin{cases} -\mu u''(x) = f(x) & x \in \Omega = (0, L), \\ u(0) = u(L) = 0, \end{cases}$$

where the diffusion coefficient $\mu \in \mathbb{R}$ is constant and such that $\mu > 0$.

We proceed by writing the weak formulation of the problem by choosing a test function v (for now without specifying its regularity), such that $v(0) = v(L) = 0$. By multiplying both sides of the strong equation by v and by integration in $[0, L]$ we have

$$-\int_0^L \mu u'' v \, dx = \int_0^L f v \, dx.$$

By partial integration as in Eq. (2.2), we have

$$\int_0^L \mu u' v' \, dx - [\mu u'(x) v(x)]_{x=0}^L = \int_0^L f v \, dx,$$

for all test functions v . The function space wherein the test function v is chosen must ensure that all the integrals are well-defined, that is the integrated function belongs to the Banach space $L^1(\Omega)$. Thus, we need to ensure that $(u' v') \in L^1(\Omega)$, for which we ask that u' and $v' \in L^2(\Omega)$. Indeed, from the Cauchy–Schwarz inequality of Corollary ?? and Theorem 2.1, we have:

$$\|u' v'\|_{L^1(\Omega)} \leq \|u'\|_{L^2(\Omega)} \|v'\|_{L^2(\Omega)}.$$

By choosing u and $v \in H^1(\Omega)$, we have that u and $v \in L^2(\Omega)$ and also u' and $v' \in L^2(\Omega)$. However, our problem is endowed with homogeneous Dirichlet boundary conditions $u(0) = u(L) = 0$; similarly, we ask that $v(0) = v(L) = 0$. For these reasons, we choose:

$$u, v \in V = H_0^1(\Omega) := \{w \in H^1(\Omega) : w(0) = w(L) = 0\}.$$

We remind that the space $H_0^1(\Omega)$ is a Hilbert space endowed with norm $H^1(\Omega)$, i.e.

$$\|w\|_{H^1(\Omega)} = \left(\|w\|_{L^2(\Omega)}^2 + \|w'\|_{L^2(\Omega)}^2 \right)^{1/2} = \left(\|w\|_{L^2(\Omega)}^2 + |w|_{H^1(\Omega)}^2 \right)^{1/2}.$$

For the functional $F(v)$, we require that the data f is “sufficiently” regular. In this case, $f \in L^2(\Omega)$ guarantees that

$$\int_0^L f v \, dx = (f, v)_{L^2(\Omega)},$$

and thus that $(f v) \in L^1(\Omega)$ thanks to the Cauchy–Schwarz inequality.

The weak formulation of the problem (WP) is:

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V,$$

where: $V = H_0^1(\Omega)$; $a : V \times V \rightarrow \mathbb{R}$ reads

$$a(u, v) := \mu \int_0^L u' v' dx$$

and $F : V \rightarrow \mathbb{R}$ is

$$F(v) := \int_0^L f v dx.$$

We use the Lax–Milgram Theorem 2.5 to show that the solution $u \in V$ of the problem in weak formulation (WP) exists and is unique. We start by recalling that the function space $V = H_0^1(\Omega)$ is a Hilbert space. Now, we proceed by verifying that the hypothesis i)–v) of the theorem are satisfied.

- i) $a : V \times V \rightarrow \mathbb{R}$ is a bilinear form. We show that $a(\eta z + \lambda v, w) = \eta a(z, w) + \lambda a(v, w)$ for any $\eta, \lambda \in \mathbb{R}$ and for any $v, w, z \in V$. We have:

$$\begin{aligned} a(\eta z + \lambda v, w) &= \int_0^L \mu (\eta z + \lambda v)' w' dx \\ &= \eta \int_0^L \mu z' w' dx + \lambda \int_0^L \mu v' w' dx = \eta a(z, w) + \lambda a(v, w), \end{aligned}$$

in virtue of linearity properties of integration and derivation. In this case, we also have $a(w, \eta z + \lambda v) = \eta a(w, z) + \lambda a(w, v)$ for all $\eta, \lambda \in \mathbb{R}$ and for all $v, w, z \in V$: therefore, $a(\cdot, \cdot)$ is a symmetric form.

- ii) $a : V \times V \rightarrow \mathbb{R}$ is a continuous form. We show that $|a(u, v)| \leq M \|u\|_V \|v\|_V$ for all $u, v \in V$, with the constat $M > 0$. We have:

$$\begin{aligned} |a(u, v)| &= \left| \mu \int_0^L u' v' dx \right| \leq \mu \int_0^L |u' v'| dx \\ &\leq \mu \|u'\|_{L^2(\Omega)} \|v'\|_{L^2(\Omega)} \leq \mu \|u\|_V \|v\|_V \quad \text{for all } u, v \in V, \end{aligned}$$

with $M = \mu$; we used the fact that $\|v'\|_{L^2(\Omega)} \leq \|v\|_{H^1(\Omega)}$ for all $v \in H^1(\Omega)$ and thus for all $v \in V$.

- iii) $a : V \times V \rightarrow \mathbb{R}$ is a coercive form. We show that $a(v, v) \geq \alpha \|v\|_V^2$ for all $v \in V$, with coercivity constant $\alpha > 0$. We have:

$$a(v, v) = \mu \int_0^L (v')^2 dx = \mu |v|_{H^1(\Omega)}^2.$$

As $v \in V = H_0^1(\Omega)$, the Poincarè inequality 2.9 holds, that is $|v|_{H^1(\Omega)}^2 \geq \frac{1}{1 + C_\Omega^2} \|v\|_{H^1(\Omega)}^2$ for all

$v \in H_0^1(\Omega)$, with $C_\Omega = \frac{L}{\sqrt{2}}$. Therefore, we have:

$$a(v, v) = \mu |v|_{H^1(\Omega)}^2 \geq \frac{\mu}{1 + C_\Omega^2} \|v\|_{H^1(\Omega)}^2 = \alpha \|v\|_V^2 \quad \text{for all } v \in V,$$

with $\alpha = \frac{\mu}{1 + C_\Omega^2} = \frac{\mu}{1 + L^2/2} > 0$. Thus, the form $a : V \times V \rightarrow \mathbb{R}$ is coercive.

iv) $F : V \rightarrow \mathbb{R}$ is a linear functional. We show that $F(\beta v + \gamma w) = \beta F(v) + \gamma F(w)$ for all $\beta, \gamma \in \mathbb{R}$ and for all $v, w \in V$. We have:

$$F(\beta v + \gamma w) = \int_0^L f(\beta v + \gamma w) dx = \beta \int_0^L f v dx + \gamma \int_0^L f w dx = \beta F(v) + \gamma F(w).$$

Therefore, $F : V \rightarrow \mathbb{R}$ is linear.

v) $F : V \rightarrow \mathbb{R}$ is continuous. As F is linear, this is continuous if also bounded. We show the last property by finding a constant $C > 0$ such that $|F(v)| \leq C \|v\|_V$ for all $v \in V$. We assume that $f \in L^2(\Omega)$ and in virtue of the Cauchy–Schwarz inequality, we have:

$$|F(v)| \leq \int_0^L |f v| dx \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)} = C \|v\|_V \quad \text{for all } v \in V,$$

where $C = \|f\|_{L^2(\Omega)}$. Thus, F is bounded and continuous.

As all the hypothesis i)–v) of the Lax–Milgram are satisfied, the solution $u \in V$ of the weak problem exists and is unique. The following stability property holds:

$$\|u\|_V \leq \frac{1}{\alpha} \|F\|_{V'} = \frac{1 + C_\Omega^2}{\mu} \|f\|_{L^2(\Omega)} = \frac{1 + L^2/2}{\mu} \|f\|_{L^2(\Omega)}.$$

Poisson problem with non homogeneous Dirichlet boundary conditions

Let us consider the following Poisson problem with the Dirichlet boundary conditions that are now non homogeneous:

$$\begin{cases} -\mu u''(x) = f(x) & x \in \Omega = (0, L), \\ u(0) = g_1, \\ u(L) = g_2, \end{cases}$$

where $g_1, g_2 \neq 0$; for simplicity, $\mu \in \mathbb{R}$ is a positive constant.

Let us introduce a function $R_g(x)$, called *lifting function* of the Dirichlet boundary data, that is $R_g \in H^1(\Omega)$, such that $R_g(0) = g_1$ and $R_g(L) = g_2$. A possible, simple choice for this function in 1D is $R_g(x) = g_1 + (g_2 - g_1)x/L$; however, infinite choices are possible. We set

$$u(x) = u_0(x) + R_g(x),$$

and we look for the weak solution u in the space $H^1(\Omega)$ as u is not zero at the boundaries of $\Omega = (0, L)$. As both R_g and $u \in H^1(\Omega)$, also $u_0 = (u - R_g) \in H^1(\Omega)$. Moreover, as $u(0) = R_g(0) = g_1$ and $u(L) = R_g(L) = g_2$, we have $u_0(0) = u_0(L) = 0$, that is $u_0 \in H_0^1(\Omega)$, which is called *homogeneous solution*.

Thus, we rewrite the problem in terms of the homogeneous solution $u_0 \in V = H_0^1(\Omega)$, wherein we also select the test function v . After our selection of the lifting function R_g and after having solved the weak problem in $u_0 \in V$, we can recover $u(x)$ as $u(x) = u_0(x) + R_g(x)$.

From the strong formulation of the problem and for $v \in V$, we have:

$$-\int_0^L \mu u'' v dx = \int_0^L f v dx;$$

by partial integration of the left hand side, we obtain:

$$\int_0^L \mu u' v' dx - \mu [u'(x) v(x)]_{x=0}^L = \int_0^L f v dx,$$

for all $v \in V$. As $V = H_0^1(\Omega)$, we have that $v(0) = v(L) = 0$ and

$$\int_0^L \mu u' v' dx = \int_0^L f v dx.$$

By setting $u = u_0 + R_g$, we have:

$$\int_0^L \mu (u_0 + R_g)' v' dx = \int_0^L f v dx \quad \text{and} \quad \int_0^L \mu u_0' v' dx = \int_0^L f v dx - \int_0^L \mu R_g' v' dx.$$

The weak formulation of the problem (WP) is:

$$\text{find } u_0 \in V : a(u_0, v) = F(v) \quad \text{for all } v \in V,$$

where $V = H_0^1(\Omega)$ and

$$a : V \times V \rightarrow \mathbb{R} : a(u_0, v) = \int_0^L \mu u_0' v' dx,$$

$$F : V \rightarrow \mathbb{R} : F(v) = \int_0^L f v dx - \int_0^L \mu R_g' v' dx = \int_0^L f v dx - a(R_g, v).$$

All the hypothesis of the Lax–Milgram theorem are satisfied, hence there exists a unique solution. In particular, $a(\cdot, \cdot)$ is a bilinear, continuous, and coercive form; $F(\cdot)$ is linear too as sum of two linear functionals ($a(\cdot, \cdot)$ is bilinear, that is linear in the second argument v , from which $a(R_g, v)$ is a linear functional for all $R_g \in H^1(\Omega)$). Moreover, F is continuous. Indeed, we have:

$$|F(v)| \leq \left| \int_0^L f v dx \right| + |a(R_g, v)| \leq (\|f\|_{L^2(\Omega)} + M\|R_g\|_{H^1(\Omega)}) \|v\|_V \quad \text{for all } v \in V,$$

with M the continuity constant of $a(\cdot, \cdot)$.

Advection–diffusion–reaction problem with homogeneous Dirichlet boundary conditions

Let us consider the problem of Sec. 2.1.2, in the domain $\Omega = (0, L)$:

$$\begin{cases} -(\mu(x) u'(x))' + \beta(x) u'(x) + \sigma(x) u(x) = f(x) & x \in \Omega = (0, L), \\ u(0) = u(L) = 0. \end{cases}$$

In general, μ , β , and σ are functions that represent the diffusion, transport, and reaction coefficients, respectively.

We choose u and $v \in V = H_0^1(\Omega)$ and, by partial integration of the diffusion term, we obtain the problem in weak formulation (WP):

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V,$$

where $a : V \times V \rightarrow \mathbb{R}$ is the following bilinear form:

$$a(u, v) = \int_0^L \mu u' v' dx + \int_0^L \beta u' v dx + \int_0^L \sigma u v dx,$$

while $F : V \rightarrow \mathbb{R}$ is the functional $F(v) = \int_0^L f v dx$. We remark that the bilinear form $a(\cdot, \cdot)$ is not symmetric in virtue of the advection term.

To give sense to the weak formulation, we need $(\mu u' v') \in L^1(\Omega)$: as $(u' v') \in L^1(\Omega)$ for u and $v \in V$, we choose $\mu \in L^\infty(\Omega)$. Indeed, from the Hölder inequality (??), we have:

$$\int_0^L |\mu(u' v')| dx \leq \|\mu\|_{L^\infty(\Omega)} \|u' v'\|_{L^1(\Omega)}.$$

Similarly, in order to have $(\beta u' v) \in L^1(\Omega)$, as $(u' v) \in L^1(\Omega)$ too for $u, v \in V$, we can choose $\beta \in L^\infty(\Omega)$. For the reaction coefficient, we can similarly select $\sigma \in L^\infty(\Omega)$ to guarantee that $(\sigma u v) \in L^1(\Omega)$. We obtain the following regularity conditions for the data:

$$\mu, \beta, \sigma \in L^\infty(\Omega) \quad \text{and} \quad f \in L^2(\Omega).$$

By applying the Lax–Milgram Theorem, it is possible to show that the weak solution $u \in V$ exists and is unique under the following additional conditions on the data:

$$\beta' \in L^\infty(\Omega) \quad \text{and} \quad \mu(x) \geq \mu_0 > 0, \quad \sigma(x) - \frac{1}{2}\beta'(x) \geq 0 \quad \text{in } \Omega.$$

Diffusion–reaction problem with Neumann boundary conditions

We consider the following diffusion–reaction problem with coefficients depending on x and Neumann boundary conditions:

$$\begin{cases} -(\mu(x) u'(x))' + \sigma(x) u(x) = f(x) & x \in \Omega = (0, L), \\ -(\mu u')(0) = q_1, \\ (\mu u')(L) = q_2, \end{cases}$$

where $q_1, q_2 \in \mathbb{R}$ and $\mu(x) \geq \mu_0 > 0$, $\sigma(x) \geq \sigma_0 > 0$. We assume that μ and $\sigma \in L^\infty(\Omega)$. As the Neumann boundary conditions do not set any fixed value for the solution at the boundaries, it is not legitimate to require that the test function v is null at the boundaries. Simply by selecting $v \in V = H^1(\Omega)$ and by partial integration, we have:

$$\int_0^L \mu u' v' dx - [(\mu u)'(x) v(x)]_{x=0}^L + \int_0^L \sigma u v dx = \int_0^L f v dx \quad \text{for all } v \in V.$$

From the Neumann data at the boundary, we have $-(\mu u)'(0) = q_1$ and $(\mu u)'(L) = q_2$, from which:

$$\int_0^L \mu u' v' dx + \int_0^L \sigma u v dx = \int_0^L f v dx + q_1 v(0) + q_2 v(L) \quad \text{for all } v \in V.$$

The weak formulation of the problem reads:

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V,$$

where $V = H^1(\Omega)$,

$$a : V \times V \rightarrow \mathbb{R}, \quad a(u, v) = \int_0^L \mu u' v' dx + \int_0^L \sigma u v dx,$$

$$F : V \rightarrow \mathbb{R}, \quad F(v) = \int_0^L f v dx + q_1 v(0) + q_2 v(L).$$

We show that the weak solution $u \in V$ exists and is unique by using the Lax–Milgram Theorem. We recall again that $V = H^1(\Omega)$ is a Hilbert space and

i) $a : V \times V \rightarrow \mathbb{R}$ is bilinear.

ii) $a : V \times V \rightarrow \mathbb{R}$ is continuous; indeed, we have

$$|a(u, v)| \leq M \|u\|_V \|v\|_V \quad \text{for all } u, v \in V,$$

with $M = \|\mu\|_{L^\infty(\Omega)} + \|\sigma\|_{L^\infty(\Omega)}$.

iii) $a : V \times V \rightarrow \mathbb{R}$ is coercive. We show that $a(v, v) \geq \alpha \|v\|_V^2$ for all $v \in V$, with $\alpha > 0$. We have:

$$\begin{aligned} a(v, v) &\geq \mu_0 \int_0^L (v')^2 dx + \sigma_0 \int_0^L v^2 dx \\ &= \mu_0 \|v\|_{H^1(\Omega)}^2 + \sigma_0 \|v\|_{L^2(\Omega)}^2 \quad \text{for all } v \in V. \end{aligned}$$

Thus:

$$a(v, v) \geq \min\{\mu_0, \sigma_0\} \|v\|_V^2 = \alpha \|v\|_V^2 \quad \text{for all } v \in V,$$

from the definition of H^1 norm, with $\alpha = \min\{\mu_0, \sigma_0\} > 0$, for $\mu_0 > 0$ and $\sigma_0 > 0$.

iv) $F : V \rightarrow \mathbb{R}$ is linear. Indeed, $F(\beta v + \gamma w) = \beta F(v) + \gamma F(w)$ for all $\beta, \gamma \in \mathbb{R}$ and for all $v, w \in V$; we have:

$$\begin{aligned} F(\beta v + \gamma w) &= \int_0^L f(\beta v + \gamma w) dx + q_1(\beta v(0) + \gamma w(0)) + q_2(\beta v(L) + \gamma w(L)) \\ &= \beta (q_1 v(0) + q_2 v(L)) + \gamma (q_1 w(0) + q_2 w(L)) = \beta F(v) + \gamma F(w). \end{aligned}$$

v) $F : V \rightarrow \mathbb{R}$ is continuous. As F a linear functional, then it is continuous if bounded. We show that F is bounded, that is there exists a constant $C > 0$ such that $|F(v)| \leq C \|v\|_V$ for all $v \in V$. As $v \in V = H^1(\Omega)$, we can use the trace inequality (2.8), for which there exists a constant $\tilde{C}_T > 0$ such that:

$$|v(0)| \leq \tilde{C}_T \|v\|_{H^1(\Omega)} \quad \text{and} \quad |v(L)| \leq \tilde{C}_T \|v\|_{H^1(\Omega)} \quad \text{for all } v \in V = H^1(\Omega).$$

Then, we obtain:

$$|F(v)| \leq \left| \int_0^L f v dx \right| + |q_1| |v(0)| + |q_2| |v(L)| \leq \|f\|_{L^2(\Omega)} \|v\|_V + \tilde{C}_T (|q_1| + |q_2|) \|v\|_V$$

for all $v \in V = H^1(\Omega)$, that is

$$|F(v)| \leq \left(\|f\|_{L^2(\Omega)} + \tilde{C}_T (|q_1| + |q_2|) \right) \|v\|_V = C \|v\|_V \quad \text{for all } v \in V = H^1(\Omega),$$

where $C = \|f\|_{L^2(\Omega)} + \tilde{C}_T (|q_1| + |q_2|)$. Thus F is bounded and continuous.

As we verified all the hypothesis of the Lax–Milgram Theorem, the weak solution $u \in V$ of the weak problem exists and is unique.

Poisson problem with mixed Dirichlet/Robin boundary conditions

For this Poisson problem, we set a homogeneous boundary condition in $x = 0$ and a Robin boundary condition in $x = L$. The problem reads:

$$\begin{cases} -\mu u''(x) = f(x) & x \in \Omega = (0, L), \\ u(0) = 0 \\ \mu u'(L) + \gamma u(L) = q, \end{cases}$$

where $\mu, \gamma \in \mathbb{R}$, with $\mu, \gamma > 0$, and $q \in \mathbb{R}$. We remark that if $\gamma = 0$, then the Robin condition coincides with the Neumann condition. As the Dirichlet condition is only in $x = 0$, we choose

$$V = H_D^1(\Omega) = \{v \in H^1(\Omega) : v(0) = 0\}.$$

By selecting $v \in V$ and by partial integration, we have:

$$\int_0^L \mu u' v' dx - [\mu u'(L) v(L) - \mu u'(0) v(0)] = \int_0^L f v dx.$$

As $v \in H_D^1(\Omega)$ and in virtue of the Robin condition $\mu u'(L) = q - \gamma u(L)$, the weak problem reads:

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V,$$

where $V = H_D^1(0, L)$,

$$a : V \times V \rightarrow \mathbb{R}, \quad a(u, v) = \int_0^L \mu u' v' dx + \gamma u(L) v(L)$$

and

$$F : V \rightarrow \mathbb{R}, \quad F(v) = \int_0^L f v dx + q v(L).$$

The former weak problem is well-posed in virtue of the Lax–Milgram Theorem.

2.3.2 PDEs in 2D and 3D

We consider as examples Poisson problems with different types of boundary conditions, for which we write the weak formulation. We consider problems in a domain $\Omega \subset \mathbb{R}^d$, with $d = 2$ or 3 .

Poisson problem with homogeneous Dirichlet boundary conditions

Let us consider again the Poisson problem (1.5) (see also Sec. 2.1.3), with homogeneous Dirichlet boundary conditions:

$$\begin{cases} -\mu \Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega; \end{cases}$$

here, $d = 2$ or 3 , and $\mu \in \mathbb{R}$, with $\mu > 0$. The domain $\Omega \subset \mathbb{R}^d$ is endowed with boundary $\partial\Omega$; $\mathbf{n}$ is the outward directed, unit vector normal to the boundary $\partial\Omega$. See for example Fig. 2.2(left) for $d = 2$.

In this case, we select the function space $V = H_0^1(\Omega)$, defined as:

$$H_0^1(\Omega) := \{v \in H^1(\Omega) : v = 0 \text{ on } \partial\Omega\},$$

for both the test function and the weak solution. Indeed, here homogeneous Dirichlet boundary conditions are used. We leverage on partial integration as in Eq. (2.5) and on the trace of test functions $v \in V = H_0^1(\Omega)$, for which we obtain the problem in weak formulation:

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V,$$

where $V = H_0^1(0, L)$,

$$a : V \times V \rightarrow \mathbb{R}, \quad a(u, v) = \int_{\Omega} \mu \nabla u \cdot \nabla v d\mathbf{x},$$

and

$$F : V \rightarrow \mathbb{R}, \quad F(v) = \int_{\Omega} f v d\mathbf{x}.$$

The former weak problem is well-posed in virtue of the Lax–Milgram Theorem.

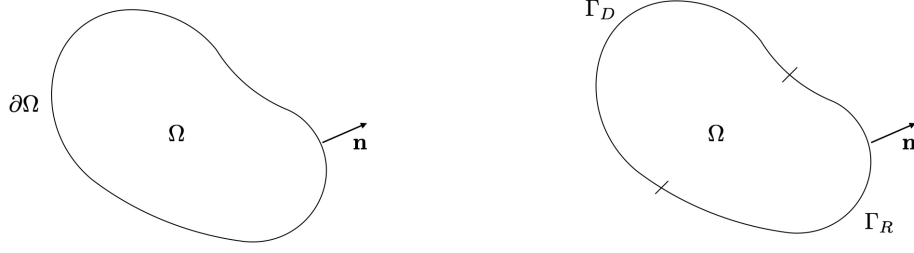


Figure 2.2: A domain $\Omega \subset \mathbb{R}^2$, its boundary $\partial\Omega$, and the outward directed, unit vector $\mathbf{n}$ normal to the boundary $\partial\Omega$ (left); boundary Ω split into the subsets Γ_D and Γ_R (right)

Poisson problem with mixed Dirichlet/Robin boundary conditions

We consider a partition of the boundary $\partial\Omega$ into two subsets Γ_D and Γ_R such that $\overset{\circ}{\Gamma}_D \cap \overset{\circ}{\Gamma}_R = \emptyset$ and $\bar{\Gamma}_D \cup \bar{\Gamma}_R = \partial\Omega$; see Fig. 2.2(right) for $\Omega \subset \mathbb{R}^2$. We consider the following problem:

$$\begin{cases} -\nabla \cdot (\mu(\mathbf{x}) \nabla u(\mathbf{x})) = f(\mathbf{x}) & \text{in } \Omega, \\ u(\mathbf{x}) = 0 & \text{on } \Gamma_D, \\ \mu(\mathbf{x}) \nabla u(\mathbf{x}) \cdot \mathbf{n} + \gamma(\mathbf{x}) u(\mathbf{x}) = q(\mathbf{x}) & \text{on } \Gamma_R. \end{cases}$$

Here, $\mu(\mathbf{x}) \geq \mu_0 > 0$ a.e. in Ω , while $\gamma(\mathbf{x}) \geq 0$ a.e. on Γ_R ; $q(\mathbf{x})$ is a given function. A homogeneous boundary condition is set on Γ_D , while a Robin one is set on Γ_R ; if $\gamma = 0$ on Γ_R , then the Neumann condition is obtained as particular case of the Robin condition.

We select the function space $V = H_D^1(\Omega)$, where

$$H_D^1(\Omega) := \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\},$$

for both the test function and the weak solution. Again, we leverage on partial integration as in Eq. (2.5), on the trace of test functions $v \in V = H_D^1(\Omega)$, and on the Robin condition Γ_R , for which we obtain the problem in weak formulation:

$$\text{find } u \in V : a(u, v) = F(v) \quad \text{for all } v \in V,$$

where $V = H_D^1(0, L)$,

$$a : V \times V \rightarrow \mathbb{R}, \quad a(u, v) = \int_{\Omega} \mu \nabla u \cdot \nabla v \, d\mathbf{x} + \oint_{\Gamma_R} \gamma u v \, d\mathbf{x}$$

and

$$F : V \rightarrow \mathbb{R}, \quad F(v) = \int_{\Omega} f v \, d\mathbf{x} + \oint_{\Gamma_R} q v \, d\mathbf{x}.$$

Again, the weak problem is well-posed in virtue of the Lax–Milgram Theorem.

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